Canadian Community Health Survey (CCHS)

Annual component

User guide

2021 Microdata file

August 2022

Statistics Canada Statistique Canada

Canada

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WHAT'S NEW IN THE 2021 CANADIAN COMMUNITY HEALTH SURVEY (CCHS)?

In 2021, due to the situation of the COVID-19 pandemic, interviews in person were suspended. The interviews were only conducted over the phone. The list dwelling frame was preferred to the area frame usually used. For this reason, the weighting adjustments in 2021 differ for the population 18 and over.

Also note that the collection of the 2021 CCHS was interrupted in the month of May for the Canadian Census of Population. It was initially planned to have six two-month collection periods. Due to operational constraints related to the 2021 Census, the collection period originally scheduled from May 1st to June 30th was rescheduled to be collected between June 1st and September 5th. The final two collection periods also have been rescheduled to be collected September 1st to November 14th and November 15th to February 7th. These changes were made in order to meet targeted response rates.

Content

The following are some of the modifications that were made to the Canadian Community Health Survey (CCHS) – Annual component questionnaire in 2021.

- The module **COVID-19 (COV)**, with questions related to the COVID-19 pandemic, was updated in 2021 to ask about vaccinations, and then was further updated in September 2021 to get more details on the number of vaccines received and the reasons for not getting vaccines. As of September 2021, questions were attributed to new modules called **COV2** and **COV3**. However, since not all questions were asked for the whole year, only those that were consistent are available on the microdata files.
- The core content module **Chronic conditions (CCC)** was updated and the conditions of asthma and chronic obstructive pulmonary disorder (COPD), which are normally asked, were removed. Please see the 2021 questionnaire for the full list of conditions being asked.
- The modules related to health behaviours **Physical activities - adults 18 years and older (PAA)**, **Physical activities for youth (PAY)**, **Sedentary Behaviours (SBE)**, **Sleep (SLP)** and **Fruit and vegetable consumption (FVC)** were asked as theme modules in the provinces and territories.
- Two modules that had been added to the CCHS as part of a rapid response related to smoking and vaping, **Smoking - stages of change 2 (SCH2)** and **Tobacco products alternatives 2 (TAL2)**, were included in all provinces and territories as theme content.
- The three modules on maternal health and breastfeeding, **Maternal experiences (MEX)**, **Smoking during maternal experience (MXS),** and **Alcohol use during maternal experience (MXA)** were included as theme modules for 2021
- The other theme modules for the year were **Activities of daily living (ADL)**, **Food security (FSC)**, **Home care services (HMC)**, and **Medication use (MED)**

- Due to the pandemic and lockdowns that occurred in 2020, the optional content consultation for 2021-2022 with the provinces and territories was not completed. As such, no optional content was selected. Rather, all provinces and territories were given the optional modules **Depression (DEP)**, **Drug use (DRG)**, **Primary health care (PHC)** and **Social provisions – reduced (SPS)**.

The following are some of the modifications that were made to specific variables during the data collection for CCHS 2021:

- Due to the evolving nature of the COVID-19 pandemic, the CCHS was not able to be updated in time for the 2021 first collection period to ask about receiving a vaccine. Since the vaccine rollout had commenced, interviewers were instructed to leave a note indicating that the respondent told them they had already received a dose, if this was mentioned when the question on vaccine willingness (COV2_040) was asked. The appropriate records from the January to February collection period were set to 'Yes' for the variable COV2_038. All others in this collection period were coded as 'not stated' since the question was not asked. This may lead to an underestimate due to non-response when using this variable. Caution should be taken when analyzing this variable.

- In the first two collection periods, only respondents who indicated that home care received was not helpful in allowing them to stay at home (HMC_037) were asked why the home care services did not help (HMC_038). Based on low sample size, starting in June 2021, the flows were adjusted for HMC_038 to also include respondents who said home care was somewhat helpful in staying home (HMC_037 = 2). Based on this change, if a respondent with SAM_CP = (202101 or 202103) answered HMC_037 = 2, the variable HMC_038 was coded as "not stated". Caution should be taken when analyzing this variable.

For more information on CCHS content, please see Appendix A for a selection of optional content by provinces and territories.

Collection

The 2021 collection of the CCHS was initially planned to have six two-month periods. However, due to operational constraints related to the 2021 Census, the collection period originally scheduled from May 1st to June 30th was rescheduled to be collected between June 1st and September 5th. The final two collection periods also have been rescheduled to be collected September 1st to November 14th and November 15th to February 7th, 2022. These changes were made in order to meet targeted response rates.

Documentation

- In order to reflect the July 2016 federal government changes to the Canadian Child Tax Benefit (CCTB) we now refer to the current system Canadian Child Benefit (CCB) throughout the documentation

- Errata: Errors that are discovered in the CCHS data files and products after dissemination are communicated to users in a cumulative error log called “CCHS Errata". The errata provides details on:
    - the products affected (e.g. master or share files or PUMF);
    - the years affected;
    - suggested corrections (if possible); and
    - steps to carry out to apply the correction

Since the release of the 2020 CCHS, there have been 0 additional entries to the errata documents.

For your convenience, the errata document has an index on the first page. It lists modules affected and the years affected. To see if a particular variable is affected, follow the hyperlinks on the associated errata item numbers.

An up to date copy of the CCHS Errata can be obtained by contacting the Centre for Population Health Data at 613-951-1746 or statcan.hd-ds.statcan@canada.ca.

1. INTRODUCTION

The Canadian Community Health Survey (CCHS) is a cross-sectional survey that collects information related to health status, health care utilization and health determinants for the Canadian population. It surveys a large sample of respondents and is designed to provide reliable estimates at the health region level. The CCHS underwent a major redesign that started in 2012 and was implemented in collection in 2015. Details of this redesign are provided in Chapter 3.

The survey's objectives are as follows:

- support health surveillance programs by providing health data at the national, provincial and health region levels;
- provide a single data source for health research on small populations and rare characteristics;
- timely release of information easily accessible to a diverse community of users; and
- create a flexible survey instrument that includes a rapid response option to address emerging issues related to the health of the population.

The CCHS data is collected from persons aged 12 and over living in private dwellings in over 100 health regions covering all provinces and territories. Excluded from the sampling frame are individuals living on Indian Reserves and on Crown Lands, institutional residents (health institutions, prisons, religious institutions, convents, etc.), full-time members of the Canadian Forces, youth aged 12 to 17 living in foster homes, and residents of certain remote regions. The CCHS covers approximately 98% of the Canadian population aged 12 and over.

The purpose of this document is to facilitate the manipulation of the CCHS microdata files and to describe the methodology used. The CCHS produces three types of microdata files: master files, share files and public use microdata files (PUMF). The characteristics of each of these files are presented in this guide. The PUMF is released after two years of data collection and contains two years of data. The next PUMF file (release date to be determined) will include the data collected for the years 2021 and 2022.

Any questions about the data sets or their use should be directed to:

Survey Help Line:	1-800-949-9491
For custom tabulations or general data support:	
Client Custom Services, Centre for Population Health Data:	613-951-1746
E-mail:	statcan.HD-DS.statcan@canada.ca
For remote access support:	613-951-1746
E-mail:	statcan.cchssm-esccds.statcan@canada.ca

2. BACKGROUND

In 1991, the National Task Force on Health Information cited a number of issues and problems with the health information system. The members felt that data was fragmented; incomplete, could not be easily shared, was not being analysed to the fullest extent, and the results of research were not consistently reaching Canadians.[1]

In responding to these issues, the Canadian Institute for Health Information (CIHI), Statistics Canada and Health Canada joined forces to create a Health Information Roadmap in 1999. From this mandate, the Canadian Community Health Survey (CCHS) was conceived. The format, content and objectives of the CCHS evolved through extensive consultation with key experts and federal, provincial and community health region stakeholders to determine their data requirements.[2] The survey started data collection in 2000.

To meet many data requirements, the CCHS had a two-year data collection cycle. Up to 2007, the first year of the survey cycle, designated by ".1", was a general population health survey, designed to provide reliable estimates at the health region level. The second year of the survey cycle, designated by ".2", had a smaller sample and was designed to provide provincial level results focused on specific health topics.

Starting in 2007, the regional component of the CCHS program began collection on an ongoing basis. The x.1 cycles of the CCHS were renamed "the annual component" of the CCHS and are now collected every year. The full title of a given annual cycle is "The Canadian Community Health Survey – Annual component, 20XX" and the short title is simply "CCHS –20XX".

The focused content component of the survey have been designated by the name of the survey followed by the topic of the themes covered by each survey (e.g., “Canadian Community Health Survey on Healthy Aging” or “CCHS – Healthy Aging”).

In 2012, a major redesign began which changed the survey sampling and collection strategy. Survey content was qualitatively tested to ensure validity and reliability in preparation for a new 8 year content plan beginning in 2015.

3. CCHS REDESIGN IN 2015

In 2012, the redesign began with extensive consultations across Canada with key experts and federal, provincial and health region stakeholders. The goal was to gather and input proposed changes and collect detailed information on the data requirements and products of the various partners.

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[1] Health Canada; Statistics Canada. 1999. *Health Information Roadmap: Responding to Needs.* p. 3.
[2] Canadian Institute for Health Information; Statistics Canada. 1999. *Health Information Roadmap: Beginning the Journey*. ISBN 1-895581-70-2. p. 19.

Below are the main changes resulting from the CCHS redesign:

- Content changes were made to most modules, but mainly to the core and theme modules. Roughly 70% of the modules that existed before the redesign have seen some level of change. Those changes vary, and have involved either minor tweaks or major changes to concepts, vocabulary, or response categories. Some new modules were created to allow for emerging concepts or data gaps identified by data sharing partners such as Health Canada, the Public Health Agency of Canada and the provincial and territorial ministries of health.

- Variable conventions have changed in 2015. The naming convention used for variables in the 2015 CCHS use a maximum of eight characters. As before, positions 1 to 3 make up the module or questionnaire section name, while position 4 indicates the variable type (underscore, D, F, C or G). Survey elements in the questionnaire (e.g. E, C, D, Q, N) now follow a uniform convention that increases in increments of five (e.g. CCC_005, CCC_010, CCC_015).

- All the derived variables were reviewed and specifications were updated with the new variable names. Content and conceptual changes were also reviewed and new derived variables were created for new content modules. The annual data collection strategy changed for 2015. Prior to 2015, there were six two-month collection periods. Starting in 2015, survey collection was divided into four non-overlapping three-month periods.

- Starting in 2015, changes to data collection in the North resulted in approximately half of the communities in each territory visited annually. As a result, the data is only representative in the territories after two years.

4. CONTENT STRUCTURE OF THE CCHS

In addition to sociodemographic and administrative data, the CCHS includes three content components, each of which addresses a different need: Common content (Core content and Theme content), Optional content, and Rapid Response content. Appendix A lists the modules included in the 2021 questionnaire and their type of component.

The average length of a CCHS interview is estimated around 40 minutes.

Table 4.1 Length of survey by component

CCHS COMPONENT		Length
Common content	Core content	**19 minutes**
	Theme content	**8 minutes**
Optional content		**5 minutes**
Rapid Response content (cost recovery basis)		**2 minutes**

4.1 Core content

The CCHS core content component includes questions asked of respondents in all provinces and territories (unless otherwise specified). These questions will remain relatively stable in the questionnaire for a period of about six years up until 2021.

4.2 Theme content

The theme content is asked of all survey respondents and comprises modules, or in some cases groups of questions within modules, that are related to a specific topic. There are two types of theme collected annually: two-year theme modules which are asked over a period of two years and one-year theme modules that are asked over a one year period. The one-year theme changes annually. Combined, the one-year and two-year theme content takes about 8 minutes of interview time. Themes are reintroduced in the survey every two, four or six years, if required. This component enables CCHS to better plan its content in the medium term.

4.3 Optional content

The optional content component is designed to give provinces and territories the opportunity to select content that addresses their own public health priorities. Each province and territory selects modules for up to 5 minutes of content. Survey participants will be asked the optional content modules that are selected for the province or territory in which they live. Optional content is selected two years at a time. It should be noted that, unlike the modules included in the core or theme content, the resulting data from the optional content modules should not be generalized across Canada[3].

Note: Due to the pandemic and lockdowns that occurred in 2020, the optional content consultation for 2021-2022 with the provinces and territories was not completed. As such, no optional content was selected. Rather, all provinces and territories were given the optional modules Depression (DEP), Drug use (DRG), Primary health care (PHC) and Social provisions – reduced (SPS).

Appendix A presents the content availability for 2021 by province or territory of residence.

4.4 Rapid response content

The rapid response component is offered on a cost-recovery basis to organizations interested in obtaining national estimates on an emerging or specific topic related to the health of the population. The rapid response content takes a maximum of two minutes of interview time. The questions appear in the questionnaire for one or two collection periods (three to six months) and are asked of all CCHS respondents during that period, excluding the Territories.

[3] Unless all provinces and territories in Canada select an optional module in the same collection period, which has never happened to date.

4.5 Content included in data files

The survey produces different data files:

- One-year data file
- Two-year combined data file
- Rapid response data file (when applicable)

Table 4.2 provides clarification about the data files available for the 2021 CCHS.

One-year data files

The survey produces data files every year. The 2021 annual file includes respondents from the 2021 data collection and variables from the core and theme content, as well as optional content. The territories are excluded from the one-year data file as the sample is only representative of the territories after two years.

Two-year data files

With each release of an even year data file, a file combining two years of data is released. The next two year file is scheduled to be released in 2023, and will include both the 2021 and 2022 reference years of collection.

The two-year data file includes all respondents and questions that were in the survey over the two year reference period. Unless otherwise specified, it comprises the questions from the annual core content and two-year theme content, as well as optional content that was selected over the two year period. The one-year theme and optional content selected for only one year are not available in the two-year data file.

Table 4.2 Content components for the 2021 and 2022 data files

Files		Annual core content	2021 one year theme content	2022 one year theme content	2021-2022 two-year theme content	Optional content[4]
2021	**Main**	Yes	Yes[5]	N/A	Yes[5]	Yes
2022	**Main**	Yes	N/A	Yes	Yes	Yes
2021 - 2022	**Main**	Yes	No	No	Yes	Yes

[4] Optional content will be included in the 2021-2022 data file if it is asked of respondents in a province or territory during the two year period. Otherwise, it will only be included in the file of the year in which it was collected. Note that if a one-year theme content module is selected for the optional content of a jurisdiction during the second year, the module will be included in the two-year data file and will be processed as optional content.

[5] One-year theme content is not available for the territories unless they have selected that module in their optional content selection for the other year.

5. SAMPLE DESIGN

5.1 Target population

The CCHS covers the population 12 years of age and over living in the ten provinces. The three territories are covered over two years of collection. Excluded from the survey's coverage are: persons living on reserves and other Aboriginal settlements in the provinces; full-time members of the Canadian Forces; youth aged 12 to 17 living in foster homes; the institutionalized population; and persons living in the Quebec health regions of Nunavik and Terres-Cries-de-la-Baie-James. Altogether, these exclusions represent less than 3% of the target population.

The coverage in the territories was exceptionally limited to the capitals (Whitehorse, Yellowknife and Iqaluit) given that in-person interviews were not possible due to the COVID-19 pandemic.

5.2 Health regions

For administrative purposes, each province is divided into health regions (HR) and each territory is designated as a single HR. Statistics Canada is sometimes asked to make minor changes to the boundaries of some of the HRs to correspond to the geography of the Census, or to better account for the health data needs determined by the new geographic boundaries. For CCHS 2021, data was collected in over 100 HRs in the ten provinces, as well as in one HR per territory (Appendix B).

5.3 Frames, household sampling strategies

The CCHS sample is selected using different frames according to the age group. For the adult population (18 years and older), the sample of households is selected from a list frame, the Dwelling Universe File (DUF). For the youth population (12 to 17 years old), a list frame, the Canada Child Benefit (CCB) file, is also used to select persons.

5.3.1 Sampling of households from the list frame of dwellings for the adult population

The DUF is a register of dwellings maintained by Statistics Canada that is primarily used to select samples. It contains a list of all residential addresses from all provinces and territories, private and collective. It is updated quarterly, using the latest administrative files available and the results of field listing and verification. Each address is geographically coded to the smallest geography level possible, which is generally at the block face level.

Up until 2020, the CCHS has always been selecting its sample of adults from an area frame in order to plan face-to-face interviews more efficiently. Selecting dwellings by simple random sampling would have led to a sample that would be too geographically spread out.

Due to the lack of face-to-face collection that continued throughout all of 2021, selecting the entire sample of dwellings from the DUF had to be considered.

Advantages in using the DUF over an area frame include the relative efficiency of the stratified simple random sample design that could be implemented (as compared to the multistage clustered

sample design selected from the area frame), as well as the simplification of many aspects of the weighting and variance estimation steps.

It was decided that the advantages for selecting an adult sample using the DUF for 2021 outweighed the advantages for selecting an adult sample from the area frame. As a result, the DUF was chosen as the frame for CCHS 2021.

The frame was stratified by HR, except in two provinces. In Ontario, strata are intersections of HRs and Local Health Integration Networks. In Alberta, strata are subdivisions of HRs named subzones.

5.3.2 Sampling of persons from the CCB frame for the youth population

To sample persons for the youth population between the ages of 12 and 17 years old, the CCHS uses a list frame created from the CCB files. The CCB files contain a list of all program beneficiaries with their names, addresses and phone numbers. This list is used to select directly the youths who will be interviewed over the phone.

The frame was stratified by HR.

5.4 Sample size and allocation

Although a primary objective of the CCHS is to produce reliable estimates for each HR given the allocated budget, the quality of the estimates for certain key characteristics at the provincial level is also deemed important. Therefore, the sample allocation strategy takes into account the needs at the provincial level. To limit response burden, a maximum sampling fraction of 10% was imposed to avoid sampling too many dwellings in smaller HRs also targeted by other surveys.

For the adult sample in the ten provinces, an initial sample of about 32,000 respondents was allocated to the provinces, using a 0.75 power allocation based on the population size. Additional sample was allocated to the provinces for which this initial allocation was insufficient to achieve the annual precision target, resulting in a sample of about 41,000 respondents. Then, within each province, the sample was allocated to the HRs using a 0.35 power allocation based on the size of the population of the HR. In Ontario and Alberta, the HR sample sizes were allocated to the strata using an allocation proportional to the size of the strata.

For the adult sample in the three territories, it was determined that an annual sample of about 240 respondents per capital would be needed to provide reliable estimates over a period of 2 years. The maximum sampling fraction of 10% limits the capacity to reach this objective in Yellowknife and Iqaluit.

The sample allocation strategy for the youth is very similar to the one developed for the adult population in the provinces. A sample of respondents was first allocated at the provincial level using a 0.75 power allocation based on the population size of the 12-17 years old. Then, within

each province, the sample was allocated at the HR level using a 0.35 power allocation also based on the population size of the 12-17 years old.

The number of respondents allocated to each strata was then inflated to compensate for expected non-response and out-of-scope units. This inflated sample size is called the raw sample size and corresponds to the number of dwellings that need to be sent to collection to obtain the targeted number of respondents.

In addition to the yearly sample, 44,000 units were added to the raw sample and sent to collection during the last two collection periods of the year. This increase was necessary to meet the needs of the Data Disaggregation Action Plan, the objective of which is to collect, analyse and disseminate more disaggregated data. The additional sample was allocated to the HRs of the ten provinces proportionally to the size of the regular raw sample, while ensuring not to exceed the maximum sampling fraction of 10%. The number of units sent to collection by province/territory for the CCHS 2021 are available in Appendix C.

5.5 Sample allocation over the collection periods

In order to balance interviewer workload and to minimize possible seasonal effects on estimates, the initial sample size for each frame is normally allocated equally over all the four collection periods. The 2021 collection of the CCHS was initially planned to have six two-month periods. However, due to operational constraints related to the 2021 Census, the collection period originally scheduled from May 1st to June 30th was rescheduled to be collected between June 1st and September 5th. The final two collection periods also have been rescheduled to be collected September 1st to November 14th and November 15th to February 7th.

Independent samples were selected in each collection period. This strategy ensures that each sample is representative of the in scope Canadian population for each period.

As mentioned in section 5.3, the raw sample in the provinces was increased by 44,000 units during the year. Approximately 25,000 were added to the penultimate collection period and 19,000 to the last collection period.

5.6 Sampling of interviewees

Since the sample for the youth population is selected from a list of individuals, the selection of the interviewee is automatically handled at sampling and no extra step is required at collection. The selection of an interviewee at collection is now only required for the adult population.

The selection strategy that was adopted accounted for user needs, cost, design efficiency, response burden and operational constraints. For a household selected as part of the 18+ sample, one person is selected per household using varying probabilities taking into account the age and the household composition. The selection probabilities resulted from simulations using various parameters in order to determine the optimal approach without causing extreme sampling weights.

Table 5.1 gives the person-level selection factors used to determine the probabilities of selection of individuals in sampled households by age group. For example, for a three-person household formed of two people between the ages of 35 and 64 and one 19-year-old, the 19-year-old would have a 1/2 chance of being selected (i.e., 20/(20+10+10)) while each of the 35 to 64 year-olds would have a 1/4 chance of being selected. To avoid extreme sampling weights, there is one exception to this rule: if the size of the household is greater than or equal to five then the person-level selection factor equals 1 for each individual in the household. Consequently, all people in that household have the same probability of being selected.

Table 5.1 Person-level selection factors, by age

Person-level selection Factors				
Age	**18-34**	**35-49**	**50-64**	**65+**
Factor	20	10	10	20

6. DATA COLLECTION

6.1 Computer-assisted interviewing

The CCHS used a computer assisted interview (CAI) application to collect data. All CCHS 2021 interviews were conducted by trained Statistics Canada interviewers over the phone using this system. During the 2021 collection year, 49,292 valid interviews were conducted in the ten provinces using CAI.

CAI offers two main advantages over other collection methods. First, CAI offers a case management system and data transmission functionality. This case management system automatically records important management information for each attempt on a case and provides reports for the management of the collection process. CAI also provides an automated call scheduler, i.e. a central system to optimise the timing of call-backs and the scheduling of appointments.

The case management system routes the questionnaire applications and sample files from Statistics Canada's main office to the interviewers laptops. Data returning to the main office takes the reverse route. To ensure confidentiality, the data is encrypted before transmission. The data are then decrypted when they are on a separate secure computer with no remote access.

Second, CAI allows for custom interviews for every respondent based on their individual characteristics and survey responses. This includes:

- questions that are not applicable to the respondent are skipped automatically
- edits to check for inconsistent answers or out-of-range responses are applied automatically and on-screen prompts are shown when an invalid entry is recorded. Immediate feedback is given to the respondent and the interviewer is able to correct any inconsistencies.

- question text, including reference periods and pronouns, is customised automatically based on factors such as the age and sex of the respondent, the date of the interview and answers to previous questions.

6.2 CCHS application development

Entry and exit components were developed to guide the interviewer through contact initiation, collection of important sample information, respondent selection and determination of case status. The health content of the application consists of the health modules themselves and made up the bulk of the application. The application underwent three stages of testing: block, integrated and end to end.

Block level testing consists of independently testing each content module or "block" to ensure skip patterns, logic flows and text, in both official languages, are specified correctly.

Integrated testing occurs when all of the tested modules are added together, along with the entry and exit components, into an integrated application. This second stage of testing ensures that key demographic information is passed from the entry to the health content and exit components of the applications. It also ensures that variables affecting skip patterns and logic flows are correctly passed between modules within the health content.

End-to-end testing occurs when the fully integrated applications are placed in a simulated collection environment. This last stage of testing allows for the testing of all technical aspects of data input, transmission and extraction of the CCHS application. It also provided a final chance of finding errors within the entry, health content and exit components.

6.3 Interviewer training

Project managers, senior interviewers and interviewers from regional collection offices were sent self-study training packages before formal training sessions and the start of collection. The focus of the training sessions were to get interviewers comfortable using the CCHS 2021 applications, and familiarise interviewers with survey content and to introduce interviewers to interviewing procedures specific to the CCHS. The training focused on:

- goals and objectives of the survey
- survey methodology
- application functionality
- review of the questionnaire content and exercises with an emphasis on significant content changes
- interviewer techniques for maintaining response - complete exercises to minimise non-response
- use of mock interviews to simulate difficult situations and practise potential non-response situations
- survey management
- transmission procedures

One of the key aspects of the training was a focus on minimizing non-response. Exercises to minimise non-response were prepared for interviewers. The purpose of these exercises was to have the interviewers practice convincing reluctant respondents to participate in the survey. There was also a series of refusal avoidance workshops given to the senior interviewers responsible for refusal conversion in each regional collection office.

6.4 The interview

Sample units were selected from a list frame, the Dwelling Universe File (DUF), for the population aged 18 and older, and the Canadian Child Benefit (CCB) frame for the population aged 12 to 17 (see section "e" below for information on youth interviews).

In all selected dwellings from the DUF (population 18 years and older), a knowledgeable adult household member was asked to supply basic demographic information on all residents of the dwelling. One member of the household was then selected for a more in-depth interview, which is referred to as the health content interview.

To ensure the quality of the data collected, interviewers were instructed to make every effort to conduct the interview with the selected respondent in privacy. In situations where this was unavoidable, the respondent was interviewed with another person present. Flags on the microdata files indicate whether somebody other than the respondent was present during the interview (ADM_N045) and whether the interviewer felt that the respondent's answers were influenced by the presence of the other person (ADM_N050).

To ensure the best possible response rate attainable, many practices were used to minimise non-response, including:

a) Introductory letters
Before the start of each collection period, introductory letters and brochures explaining the purpose of the survey were sent to the sampled households. These explained the importance of the survey and provided examples of how CCHS data would be used.

b) Initiating contact
Interviewers were instructed to make all reasonable attempts to obtain interviews. Numerous call-backs were made at different times on different days. When the timing of the interviewer's call was inconvenient, an appointment was made to call back at a more convenient time.

c) Refusal conversion
For individuals who at first refused to participate in the survey, a letter was sent from the nearest Statistics Canada Regional Office to the respondent, stressing the importance of the survey and the household's collaboration. This was followed by a second call from a senior interviewer, a project supervisor or another interviewer to try to convince respondent of the importance of participating in the survey.

d) Language barriers

To remove language as a barrier to conducting interviews, each of the Statistics Canada Regional Offices recruited interviewers with a wide range of language competencies. When necessary, cases were transferred to an interviewer with the language competency needed to complete an interview.

e) Youth interviews

Interviewers needed to obtain verbal permission from parents/guardians to interview youths between the ages of 12 to 14 who were selected for interviews. This information was collected in the Parental/Guardian Consent (PGC) block in the survey entry component. Several procedures were followed by interviewers to alleviate potential parental concerns and to ensure a completed interview.

If a parent/guardian asked to see the actual questions; interviewers were instructed to immediately have the regional office send a copy of the questionnaire.

If privacy could not be obtained to interview the selected youth over the phone (another person listening in) the interview was coded a refusal.

The Person Most Knowledgeable (PMK) block collected household level information found at the end of the survey (Insurance coverage, Food Security, Income and Administration) from the most knowledgeable person in the household. This block is initiated when the selected respondent is between the ages of 12 to 17. The block formalizes the process of identifying a person in the household who is likely better able to answer these household level questions than the young selected respondent. If a PMK is found, then the interview moves from the younger selected respondent between the ages of 12 and 17, to a household member who finishes the rest of the interview after the PMK block.

f) Proxy interviews

In cases where the selected respondent was, for reasons of physical or mental health, incapable of completing an interview, another knowledgeable member of the household supplied information about the selected respondent. This is known as a proxy interview. While proxy interviewees were able to provide accurate answers to most of the survey questions, the more sensitive or personal questions were beyond the scope of knowledge of a proxy respondent. This resulted in some questions from the proxy interview being unanswered. Every effort was taken to keep proxy interviews to a minimum.

The Proxy interview (GR) block prompts the interviewer to specifically identify whether the proxy interview is being conducted due to a physical or mental condition. Interviewers are then asked to record the specific condition for either case. The variable ADM_PRX indicates whether a case was completed by proxy.

6.5 Field operations

The CCHS 2021 was divided into five non-overlapping collection periods. Collection began January 4, 2021 and was originally scheduled to complete on December 24, 2021. Since collection for CCHS 2022 was delayed until February 2022, CCHS 2021 collection was extended to January 31, 2022 to maximise response rates which were negatively impacted by the pandemic.

Regional collection offices were instructed to use the first few weeks of each collection period to resolve the majority of the sample, and use the remaining weeks to finalise the rest of the sample and to follow up on outstanding non-response cases. All cases were to have been attempted by the second week of each collection period. Sample files were sent approximately two weeks before the start of each collection period. A series of dummy cases were included with each sample. These cases were completed by senior interviewers for the purposes of ensuring that all data transmission procedures were working through the collection cycle.

For final response rates, refer to Appendix D.

6.6 Quality control and collection management

During the collection year, several methods are used to ensure data quality and to optimize collection. These included using internal measures to verify interviewer performance and the use of a series of ongoing reports to monitor various collection targets and data quality.

A system of validation was used whereby interviewers had their work validated on a regular basis by the senior collection staff. Each collection period, randomly selected cases were flagged in the sample. Validation procedures generally occurred during the first few weeks of a collection period to ensure that any issues were detected promptly. Interviewers were provided feedback by their supervisors on a regular basis.

Responsive Collection Design (RCD) allows collection approaches to be adjusted throughout the collection period. In basic terms, it allows the grouping cases according to a number of factors, and then target the groups that are of particular interest. In an ordinary survey using the CATI collection platform, the interviewer groups remain constant throughout collection. However, during Responsive Collection Design, these groups change during each phase of collection in order to maximize response rates, productivity and the representativeness of the collected sample.

With the 2015 CCHS redesign, a dynamic telephone number strategy was developed. Starting in 2015, the 18+ years old sample can have up to 2 phone numbers, while the CCB samples can have up to 4 telephone numbers. This increases the number of available telephone numbers interviewers can call to maximize their potential ability to reach the selected households or respondents.

7. DATA PROCESSING

7.1 Editing

Most editing of the data was performed at the time of the interview by the computer-assisted interviewing (CAI) application. It was not possible for interviewers to enter out-of-range values and flow errors were controlled through programmed skip patterns. For example, CAI ensured that questions that did not apply to the respondent were not asked.

In response to some types of inconsistent or unusual reporting, warning messages were invoked but no corrective action was taken at the time of the interview. Where appropriate, edits were instead developed to be performed after data collection at Head Office. Inconsistencies were usually corrected by setting one or both of the variables in question to "not stated".

7.2 Flows

When data collected in the field is returned to Head Office, any questions that were skipped will appear as blank on the preliminary data files. During processing, every blank variable is given one of two processing code values:

‘Valid Skip’ – VS
For variables of length 1, the value for VS will simply be 6. Otherwise, these values will appear on the file as a string of 9’s, followed by a 6, according to the length of the numerical variable or text field.

The code VS is applied whenever a question is skipped because it does not apply to the respondent (a population exclusion). The two main reasons for this code being used are optional module variables for respondents in provinces where that content was not selected; and questions that are skipped by a flow in the questionnaire since it did not apply, as with the Maternal Experiences questions for male respondents.

‘Not Stated’ – NS
For a variable of length 1, the value for NS will simply be 9. Otherwise, these will appear on the file as a string of 9’s, according to the length of the numerical variable or text field.

The code NS is applied whenever a question is skipped, but could have been asked. These are respondents that are in the universe, but without an actual response. The two main reasons for this code being used are questions that are skipped because the interview is being completed by proxy; and questions that are skipped because a preceding question was answered with a ‘Don’t Know’, ‘Refuse’, or else was itself ‘Not Stated’. For example, if a respondent refuses to answer CCC_Q095 (Do you have diabetes?), the questions that ask about the age of the diagnosis, insulin use, etc., are all set to ‘Not Stated’ since it is not known whether or not the respondent has diabetes.

7.3 Coding

Pre-coded answer categories were supplied for all suitable variables. Interviewers were trained to assign the respondent's answers to the appropriate category.

In the event that a respondent's answer could not be easily assigned to an existing category, several questions also allowed the interviewer to enter a long-answer text in the "Other-specify" category. All such questions were closely examined in head office processing. For some of these questions, write-in responses were coded into one of the existing listed categories if the write-in information duplicated a listed category. In 2021, variables that were OSLA coded include GDR_010, SDC_037, CAN_025, CAN_035, and PG_01. Other variables with other-specify responses for this cycle were not selected for coding. For all questions, the 'Other-specify' responses are taken into account when refining the answer categories for future cycles.

When write-in responses ("Other-specify") are coded into an existing category during head office processing, it is possible that other questions in the questionnaire would become in scope to the respondent. Since those questions were never asked during the interview, the missing answers were set to 'Not Stated' in processing.

7.4 Creation of derived variables

To facilitate data analysis and to minimize the risk of error, a number of variables on the file have been derived using items found on the CCHS questionnaire. Derived variables generally have a "D", "G" or "F" in the fourth character of the variable name. In some cases, the derived variables are straightforward, involving the collapsing of response categories. In other cases, several variables have been combined to create a new variable. The *Derived Variables Documentation (DV)* provides details on how these more complex variables were derived. For more information on the naming convention, please go to Section 12.6.

7.5 Weighting

The principle behind estimation in a probability sample such as CCHS is that each person in the sample "represents", besides himself or herself, several other persons not in the sample. For example, in a simple random 2% sample of the population, each person in the sample represents 50 persons in the population. In the terminology used here, it can be said that each person has a weight of 50.

The weighting phase is a step that calculates, for each person, his or her associated sampling weight. This weight appears on the microdata files and must be used to derive meaningful estimates from the survey. For example, if the number of individuals who smoke daily is to be estimated, it is done by selecting the records referring to those individuals in the sample having that characteristic and summing the weights entered on those records.

7.6 Income variables for 2021 - Linkage and Imputation

Since 2016, income variables in the 2021 CCHS master file are populated from one of three sources: tax records, respondent provided data, imputed data. Where respondents did not object to a link to their tax data and where a link to the appropriate tax records could be found, the income source variables (INC_005A-O, INC_Q025A-O), the main source of income (INC_015, INC_035), if they received a supplement for people with a disability (INC_010, INC_030) and the total income amounts (INC_021, INC_041) were determined from tax records. Where linkage to tax data was not feasible, respondent data was used. If neither linked tax data nor respondent reported data was available, an imputation of the total personal and total household income variables was conducted. The sources of incomes (INC_005A-O, INC_010, INC_015, INC_025A-O, INC_030, INC_035) were not imputed.

The variable INCFIMPP indicates the source used for the personal income variables and INCFIMPH indicates the source for household income variables. Starting in 2019, any tax source will be indicated as imputed. For more information concerning income on the master file, please refer to the document "CCHS_2021_Master_File_Income" available in the 'Documentation' folder of the 2021 CCHS. For more information concerning income on the share file, please refer to the document "CCHS_2021_Share_File_Income".

7.7 Postal code imputation

A sampled unit's postal code is used together with the Postal Code Conversion File, to derive the rest of the geographical variables that are available on the CCHS data file. It is therefore important that all sampled units have a valid postal code. If a sampled unit's postal code is missing or invalid, it is usually imputed through a donor imputation process, although other imputation methods are sometimes used. The donor is chosen from the same geographical area, with as much precision as possible, as the unit with the missing or invalid postal code.

For the 2021 CCHS, approximately 4.0% of sampled units from the list dwelling frame (those aged 18 and over) had their postal codes imputed. For sampled units from the CCB frame (aged 12-17), approximately 1.2% had their postal codes imputed in 2021.

8. WEIGHTING

In order for estimates produced from survey data to be representative of the covered population, and not just the sample itself, users must incorporate the survey weights in their calculations. A survey weight is given to each respondent included in the final sample. This weight corresponds to the number of persons in the entire population that are represented by the respondent.

As described in section 5, the CCHS uses two sampling frames for its sample selection: an list dwelling frame for the Canadian population aged 18 and over and a list frame from Canada Child Benefit (CCB) records for the 12-17 population.

In 2021, due to the COVID-19 pandemic, interviews in person were suspended. The interviews were only conducted over the phone. The list dwelling frame was preferred to the area frame usually used (See Section 5.3.1 for details). For this reason, the weighting adjustments in 2021 differ for the population 18 and over.

Also note that the collection of the CCHS was interrupted in the month of May for the Canadian Census of Population.

The weighting strategy treats both the list dwelling frame and CCB frame independently to come up with separate person-level weights for each of the frames used. The person-level weights from the two frames are then combined into a single set of weights, jointly undergo a couple more adjustments (including being matched to known population totals) and become the final person-level weights.

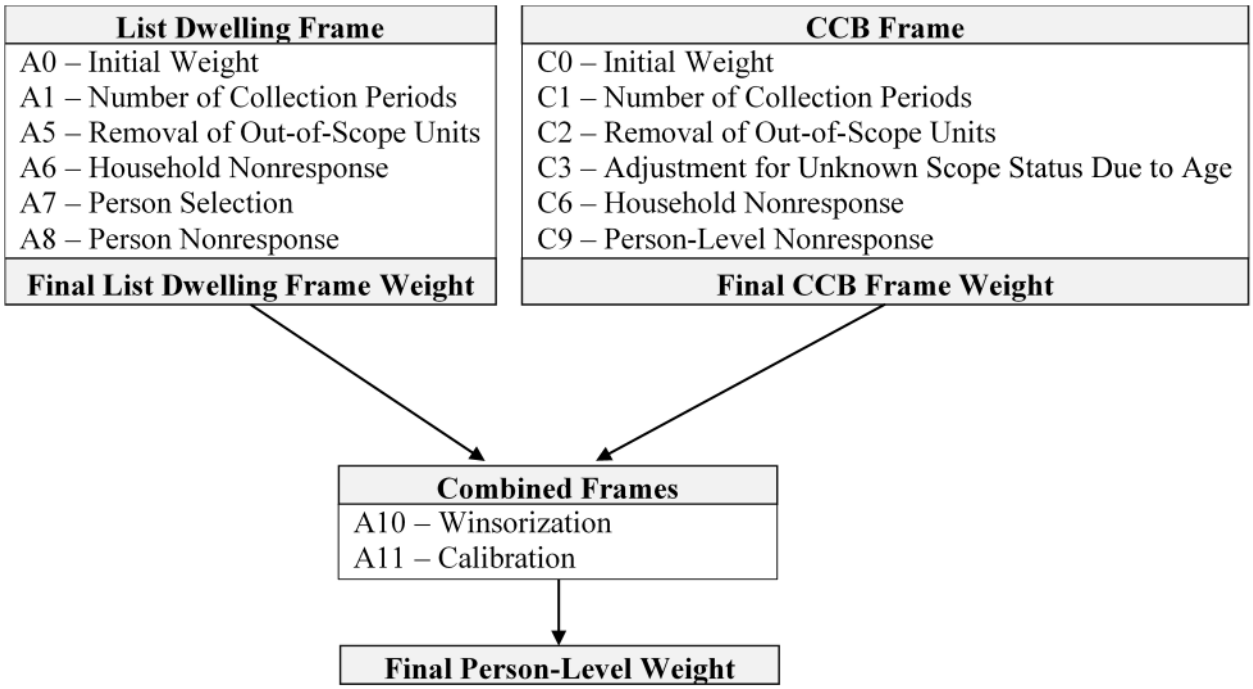
8.1 Overview

The following subsections describe the weighting process for the provinces. Units from both the list dwelling frame and the CCB frame are treated separately up to and including the adjustment for person-level nonresponse. Sub-section 8.2 provides details on the weighting strategy for the list dwelling frame, while sub-section 8.3 deals with the strategy for the CCB frame. The final weighting steps of Winsorization and calibration, which involve both frames, are discussed in sub-section 8.4.

Following the redesign of the CCHS in 2015, due to operational constraints, collection of a representative sample within some geographies are spread over a two-year period. For example, estimates in the three territories are typically representative over two years. As such, respondents in the territories are omitted from the one-year files and will only be included on the two-year files. There are some particularities associated with the weighting process for the territories, and this will be discussed in more detail in the documentation of the two-year CCHS files.

Many adjustments to unit weights are made throughout the weighting process. Adjustments applied to units from the list dwelling frame are enumerated from *A0* to *A9*, while adjustments applied to units from the CCB frame are enumerated from *C0 to C9*. The final adjustments after combining the adult and CCB frames are called *A10* and *A11*. Diagram A presents an overview of the most important of these adjustments that are part of the weighting strategy.

Diagram A - Weighting Strategy Overview



The subsections 8.2 to 8.4 present the details behind all the steps shown in Diagram A.

8.2 Weighting of the list dwelling frame sample

A0 – Initial Weight

The initial design weight is defined as the inverse of the probability of selection. For the list dwelling frame, dwellings are selected from the Dwelling Universe File (DUF) within each CCHS stratum (Health Region).

Dwellings are randomly selected from each HR. The probability of selection corresponds to the ratio of the number of sampled units to the number of dwellings on the list within the HR. The ratio is based on the frame available and the number of units selected for the particular collection period. The probability of selection can therefore change depending on sample allocation and frame updates. The inverse of these probabilities represents the initial weight A0.

A1 – Number of Collection Periods

Samples are drawn for every collection period on the list dwelling frame. Each of these samples comes with an initial weight that allows each sample to be representative of the population at the HR level. To ensure that the total sample represents the population only once, an adjustment factor is applied to reduce the weights of each collection period sample. The adjustment factor applied to each collection period sample is equal to the inverse of the number of samples being combined (i.e. the number of collection periods). Following this adjustment, the entire list frame sample

corresponds to the average population over the entire combined collection period. The initial weights (A0) are multiplied by this adjustment factor to produce weight A1.

A5 – Removal of Out-of-Scope Units

Among all dwellings sampled, a certain proportion are identified during collection as being out-of-scope. Dwellings that are demolished or under construction, vacant, seasonal or secondary, as well as institutions are examples of out-of-scope cases for the CCHS. Dwellings in which all residents are out-of-scope, for example, all are full-time members of the Armed Forces or visitors to Canada are also out-of-scope. All these out-of-scope dwellings and their associated weights are simply removed from the sample, as it is assumed that the weighted out-of-scope dwellings in the sample are representative of out-of-scope dwellings in the population. For dwellings whose scope status is unknown (not enough information was gathered to determine scope status), their weights are reduced by the out-of-scope rate of their HR to account for some proportion of them being out-of-scope. This leaves a sample that is representative of in-scope dwellings or households. In-scope dwellings maintain the same weight as in the previous step, A1, which is now called weight A5.

A6 – Household Nonresponse

During collection, a certain proportion of sampled households inevitably result in nonresponse. This usually occurs when a household refuses to participate in the survey, provides unusable data, or cannot be reached for an interview. Weights of the nonresponding households are redistributed to responding households within response homogeneity groups (RHGs), which are formed within province. In order to create the RHGs, a scoring method based on logistic regression models is used to determine the propensity to respond and these response probabilities are used to divide the sample into groups with similar response properties. The information available for non-respondents is limited so the regression models use characteristics such as the collection period, geographic information and variables from Statistics Canada's Household Survey Frame, as well as paradata or process data, which includes the number of contact attempts, the time/day of attempt, and whether the household was called on a weekend or weekday. An adjustment factor is calculated within each RHGs as follows:

$$\frac{Sum\ of\ weights\ A5\ for\ all\ households}{Sum\ of\ weights\ A5\ for\ all\ responding\ households}$$

Weight A5 is multiplied by this adjustment factor to produce weight A6 for the responding households. Non-responding households are dropped from the process at this point.

A8 - Person-level Selection

Since persons are the desired sampling units, the household-level weights computed to this point need to be converted to the person level. This weight is obtained by multiplying the household weight by the inverse of the probability of selection of the person selected in the household. This gives the weight A8. The probability of selection for an individual living in a dwelling selected from the list dwelling frame changes depending on the number of people in the household and their ages (see subsection 5.6 for more details).

A9 - Person-level Nonresponse

A CCHS interview for the adult frame dwellings can be seen as a two-part process. First, the interviewer gets the complete roster of the people within the household. Second, the selected person is interviewed. In some cases, interviewers can only get through the first part, either because they cannot get in touch with the selected person, or because that selected person refuses to be interviewed. Such individuals are defined as person nonrespondents and an adjustment factor must be applied to the weights of person respondents to account for this nonresponse. Using the same methodology that is used in the treatment of household nonresponse, the adjustment is applied within response homogeneity groups. In this process, the scoring method is used to define a response probability based on characteristics available for both respondents and nonrespondents. In addition to characteristics previously used to model household-level nonresponse (geographic information and paradata), all characteristics collected when creating the roster of household members are available for the estimation of response probabilities. The probabilities are grouped into response homogeneity groups and the following adjustment factor is calculated within each group:

$$\frac{Sum\ of\ weights\ A8\ for\ all\ selected\ persons}{Sum\ of\ weights\ A8\ for\ all\ responding\ selected\ persons}$$

Weight A8 for responding persons is multiplied by the above adjustment factor to produce weight A9. Nonresponding persons are dropped from the weighting process from this point onward.

8.3 Weighting of the CCB frame sample

C0 – Initial Weight

The initial design weight is defined as the inverse of the probability of selection. For the CCB frame, the selection of children is done from the CCB list frame within each CCHS stratum.

For the CCB frame, children are randomly selected among those assigned to the specific HR. The probability of selection corresponds to the ratio of the number of sampled units to the number of children on the list within the HR. The ratio is based on the frame available and the number of units selected for the particular collection period. The probability of selection can therefore change depending on sample allocation and frame updates. The inverse of these probabilities represents the initial weight C0.

C1 – Number of Collection Periods

Samples are drawn for every collection period on the CCB frame. Each of these samples comes with an initial weight that allows each sample to be representative of the population at the HR level. To ensure that the total sample represents the population only once, an adjustment factor is applied to reduce the weights of each collection period sample. The adjustment factor applied to each collection period sample is equal to the inverse of the number of samples being combined (i.e. the number of collection periods). Following this adjustment, the entire list frame sample corresponds to the average population over the entire combined collection period. The initial weights (C0) are multiplied by this adjustment factor to produce weight C1.

C2 – Removal of Out-of-Scope Units

Selected children who are full-time members of the Canadian Armed Forces, are part of a First Nations reserve, or are deceased, are all examples of out-of-scope cases for the CCB frame. Similar to the methods used on the list dwelling frame, these cases are simply removed from the process, leaving only in-scope children in the sample. These in-scope children keep the same weight as in the previous step, now called weight C2.

C3 – Adjustment for Unknown Scope Status Due to Age

Some of the selected children have birthdays that fall within a collection period. For a proportion of these children, their birthday can change their scope status due to age. Some selected children may be eleven at the beginning of the collection period and therefore out-of-scope, but become in-scope when they turn twelve. Other children may be in-scope at seventeen but become out-of-scope when they turn eighteen. The age of these children, and therefore scope status, can be resolved (and treated accordingly) when contact is made and the age at time of contact is established. However, the scope status is unknown when children cannot be contacted or asked their age. They have their weights adjusted by their modeled probability of being in-scope had they been contacted and asked their age at contact. The rate of in-scope children turning 12 and 18 amongst known scope cases is used as the adjustment factor for the weights of these unknown scope cases. Those found out of scope due to their age have their weights set to zero similar to other out-of-scope units, in-scope children keep their weights and those with unknown scope and birthdays that could affect their scope have their weights adjusted, the resulting weights are called C3.

C6 – Household Nonresponse

The adjustment applied here to compensate for the effect of household nonresponse is identical to the one applied for the list dwelling frame units that were contacted by telephone (adjustment A6). For the CCB frame the selection was done at the person-level. However a household response is defined as a household where the roster was completed, similar to the list dwelling frame. The adjustment factor calculated within each response homogeneity group is obtained as follows:

$$\frac{Sum\ of\ weights\ C3\ for\ all\ households}{Sum\ of\ weights\ C3\ for\ all\ responding\ \ households}$$

The weight C3 of responding households is multiplied by this adjustment factor to produce the weight C6. Nonresponding households are removed from the process at this point.

C9 – Person-level Nonresponse

The adjustment applied here to compensate for the effect of person nonresponse is identical to the one applied for the list dwelling frame (adjustment A9). The adjustment factor calculated within each response homogeneity group is obtained as follows:

$$\frac{Sum\ of\ weights\ C6\ for\ all\ persons}{Sum\ of\ weights\ C6\ for\ all\ responding\ persons}$$

The weight C6 of responding persons is multiplied by this adjustment factor to produce the weight C9. Nonresponding persons are removed from the process at this point.

8.4 Final weight adjustments with adult and CCB frames combined

At this point in the process the sample from the two frames can be combined in anticipation of the final weight adjustments of Winsorization and calibration. It is not necessary to perform an additional weight adjustment in combining the frames because they cover mutually exclusive populations. The selection probability and subsequent weighting of a unit on a given frame is in no way impacted by the structure or selection of units from the other frame.

A10 – Winsorization

Following the series of adjustments applied to the respondents, some units may come out with extreme weights compared to other units of the same domain of interest. These units could have a large impact on the variance. In order to prevent this, the weight of these outlier units is adjusted downward using a "winsorization" trimming approach.

A11 – Calibration

The last step necessary to obtain the final CCHS weight is calibration (A11). Calibration is done using Statistics Canada's generalized estimation system, G-EST, to ensure that the sum of the final weights corresponds to the population estimates defined at the HR level, for all 12 age-sex groups of interest. The six age groups are 12-17, 18-34, 35-49, 50-64, 65-74 and 75+, for both males and females. A minimum size of 20 respondents is required to calibrate at the HR by age and sex. When getting less than 20 respondents, some collapsing is done within the province and/or within sex. At the same time, weights are adjusted to ensure that each collection period is equally represented within the sample. Note that the calibration is done using the most up to date geography and may not match the geography used at the time of sampling.

Since there was no collection in the month of May, the Census projections for that month were not used in the calibration process.

The population estimates are projections based on the 2016 Census counts and counts of birth, death, immigration and emigration since that time. The average of these monthly estimates for each of the HR-age-sex post-strata by collection period is used to calibrate. The weight A10 is adjusted using G-EST to obtain the final weight A11. Weight A11 corresponds to the ***final CCHS person-level weight*** and can be found on the master file with the variable name WTS_M.

8.5 Creation of a share weight

Along with the master file which contain all CCHS respondents, a share file is created which contains only a portion (usually > 90%) of the original CCHS respondents. The individuals on this

share file have agreed to share their data with certain partners. To compensate for the loss of some respondents from the file, the weights of the respondents who agreed to share must be adjusted by the factor:

$$\frac{Sum\ of\ weights\ A9\ (or\ C9)\ for\ all\ respondents}{Sum\ of\ weights\ A9\ (or\ C9) for\ all\ respondents\ agreeing\ to\ share\ their\ data}$$

Similar to the nonresponse adjustments, this factor is calculated within response homogeneity groups, where in this case, individuals with similar estimated propensity to share will be grouped together. This share adjustment is calculated separately for respondents from the adult frame and from the CCB frame. Weight A9 (or C9 for the CCB respondents) for respondents who agreed to share is multiplied by the above adjustment factor to produce a share weight. As with the Master weighting process, the adult and CCB frames are combined at this point, and Winsorization and calibration (similar to adjustments A10 and A11) are applied to the share units. The final weight after these adjustments is called WTS_S.

9. DATA QUALITY

9.1 Response rates

In total, 206.403 of the selected units in the CCHS 2021 were in-scope for the survey[6], out of which a response was obtained for 49,678 individuals, resulting in a response rate of 24.1%. Appendix D provides the counts of in-scope units, the counts of respondents and the resulting response rates, by province/territory. These figures are provided for the adults and for the youths separately.

9.2 Survey Errors

The estimates derived from this survey are based on a sample of individuals. Somewhat different figures might have been obtained if a complete census had been taken using the same questionnaire, interviewers, supervisors, processing methods, etc. than those actually used. The difference between the estimates obtained from the sample and the results from a complete count under similar conditions is called the ***sampling error*** of the estimate.

Errors which are not related to sampling may occur at almost every phase of a survey operation. Interviewers may misunderstand instructions, respondents may make errors in answering questions, the answers may be incorrectly entered on the computer and errors may be introduced in the processing and tabulation of the data. These are all examples of ***non-sampling errors***.

[6] Among the units selected, some are not in-scope for the survey. They are, for example, dwellings selected from the list frame of dwellings but that turned out to be vacant, demolished or non-residential dwellings, or youths that have moved outside Canada. These units are identified during the data collection, otherwise they would have been excluded before the sample selection. These units are not considered in the calculation of response rates. This response rate includes the three territories. Data for the three territories will be available in the 2021-2022 CCHS.

9.2.1 Non-Sampling Errors

Over a large number of observations, randomly occurring errors will have little effect on estimates derived from the survey. However, errors occurring systematically will contribute to biases in the survey estimates. Considerable time and effort was made to reduce non-sampling errors in the CCHS. Quality assurance measures were implemented at each step of data collection and processing to monitor the quality of the data. These measures included the use of highly skilled interviewers, extensive training with respect to the survey procedures and questionnaire, and the observation of interviewers to detect problems. Testing of the CAI application and field tests were also essential procedures to ensure that data collection errors were minimized.

A major source of non-sampling errors in surveys is the effect of ***non-response*** on the survey results. The extent of non-response varies from partial non-response (failure to answer just one or some questions) to total non-response. Partial non-response to the CCHS was minimal; once the questionnaire was started, it tended to be completed with very little non-response. Total non-response occurred either because a person refused to participate in the survey or because the interviewer was unable to contact the selected person. The response rates were lower for the CCHS 2021 than for a normal cycle of the CCHS, due to the impact of the COVID-19 pandemic on the collection activities. For units that did not have a telephone number available on the sampling frame, the contact rates were very low since the usual in-person visits by interviewers were not allowed (in-person visits gradually resumed in July 2021; however, only quick interactions to present the CCHS and collect a phone number were allowed). For units that could be contacted by phone, having shorter collection periods and reduced interviewer capacity for CCHS data collection were the main reasons for the drop in response rates. Total non-response was handled by adjusting the weight of persons who responded to the survey to compensate for those who did not respond. See section 8 for details on the weight adjustment for non-response.

9.2.2 Sampling Errors

Since it is an unavoidable fact that estimates from a sample survey are subject to sampling error, sound statistical practice calls for researchers to provide users with some indication of the magnitude of this sampling error. The basis for measuring the potential size of sampling errors is the standard deviation of the estimates derived from survey results. However, because of the large variety of estimates that can be produced from a survey, the standard deviation of an estimate is usually expressed relative to the estimate to which it pertains. This resulting measure, known as the coefficient of variation (CV) of an estimate, is obtained by dividing the standard deviation of the estimate by the estimate itself and is expressed as a percentage of the estimate.

For example, suppose hypothetically that it is estimated that 25% of Canadians aged 12 and over are regular smokers and that this estimate is found to have a standard deviation of 0.003. Then the CV of the estimate is calculated as:

$$(0.003/0.25) \times 100\% = 1.20\%$$

Statistics Canada commonly uses CV results when analyzing data and urges users producing estimates from the CCHS data files to also do so. For details on how to determine CVs, see Section 11. For guidelines on how to interpret CV results, see the table at the end of sub-section 10.4.

10. GUIDELINES FOR TABULATION, ANALYSIS AND RELEASE

This section of the documentation offers guidelines to users for tabulating, analyzing, publishing or otherwise releasing any estimates derived from the survey files. With the aid of these guidelines, users of microdata should be able to produce figures that are in close agreement with those produced by Statistics Canada. They will also be able to develop currently unpublished figures in a manner consistent with these established guidelines. Methods to measure precision and quality are also described along with release guidelines to help decide when an estimate should be used in publication.

10.1 Rounding guidelines

In order that estimates for publication or other release derived from the data files (Master, Share or PUMF) correspond to those produced by Statistics Canada, users are urged to adhere to the following guidelines regarding the rounding of such estimates:

a) Estimates in the main body of a statistical table are to be rounded to the nearest hundred units using the normal rounding technique. In normal rounding, if the first or only digit to be dropped is 0 to 4, the last digit to be retained is not changed. If the first or only digit to be dropped is 5 to 9, the last digit to be retained is raised by one. For example, in normal rounding to the nearest 100, if the last two digits are between 00 and 49, they are changed to 00 and the preceding digit (the hundreds digit) is left unchanged. If the last digits are between 50 and 99 they are changed to 00 and the proceeding digit is incremented by 1;

b) Marginal sub-totals and totals in statistical tables are to be derived from their corresponding unrounded components and then are to be rounded themselves to the nearest 100 units using normal rounding[7];

c) Averages, proportions, rates and percentages are to be computed from unrounded components (i.e., numerators and/or denominators) and then are to be rounded themselves to one decimal using normal rounding. In normal rounding to a single digit, if the final or only digit to be dropped is 0 to 4, the last digit to be retained is not changed. If the first or only digit to be dropped is 5 to 9, the last digit to be retained is increased by 1;

[7] CCHS tables for 2015 onwards use an algorithm that round totals, then solves internally within the table to match the totals. Due to this method of rounding, tables generated using CCHS microdata may not replicate totals found in the data tables

d) Sums and differences of aggregates (or ratios) are to be derived from their corresponding unrounded components and then are to be rounded themselves to the nearest 100 units (or the nearest one decimal) using normal rounding;

e) In instances where, due to technical or other limitations, a rounding technique other than normal rounding is used resulting in estimates to be published or otherwise released that differ from corresponding estimates published by Statistics Canada, users are urged to note the reason for such differences in the publication or release document(s);

f) Under no circumstances are unrounded estimates to be published or otherwise released by users. Unrounded estimates imply greater precision than actually exists.

10.2 Sample weighting guidelines for tabulation

The sample design used for this survey was not self-weighting. That is to say, the sampling weights are not identical for all individuals in the sample. When producing simple estimates, including the production of ordinary statistical tables, users must apply the proper sampling weight. If proper weights are not used, the estimates derived from the data file cannot be considered to be representative of the survey population, and will not correspond to those produced by Statistics Canada.

Users should also note that some software packages might not allow the generation of estimates that exactly match those available from Statistics Canada, because of their treatment of the weight field. If options are available, users should ensure that they specify that the weight is a sample weight rather than a frequency weight.

10.2.1 Definitions: categorical estimates, quantitative estimates

Before discussing how the survey data can be tabulated and analyzed, it is useful to describe the two main types of point estimates of population characteristics that can be generated from the data files.

<u>Categorical estimates:</u>
Categorical estimates are estimates of the number or percentage of the surveyed population possessing certain characteristics or falling into some defined category. The number of individuals who smoke daily is an example of such an estimate. An estimate of the number of persons possessing a certain characteristic may also be referred to as an estimate of an aggregate.

Example of categorical question:
At the present do/does ... smoke cigarettes daily, occasionally or not at all? ***(SMK_005)***
__ Daily
__ Occasionally
__ Not at all
<u>Quantitative estimates:</u>

Quantitative estimates are estimates of totals or of means, medians and other measures of central tendency of quantities based upon some or all of the members of the surveyed population.
An example of a quantitative estimate is the average number of cigarettes smoked per day by individuals who smoke daily. The numerator is an estimate of the total number of cigarettes smoked per day by individuals who smoke daily, and its denominator is an estimate of the number of individuals who smoke daily.

Example of quantitative question:
How many cigarettes do/does you/he/she smoke each day now? ***(SMK_045)***
|_|_| Number of cigarettes

10.2.2 Tabulation of categorical estimates

Estimates of the number of people with a certain characteristic can be obtained from the data file by summing the final weights of all records possessing the characteristic of interest.

Proportions and ratios of the form $\hat{X} / \hat{Y}$ are obtained by:

a) summing the final weights of records having the characteristic of interest for the numerator ($\hat{X}$);
b) summing the final weights of records having the characteristic of interest for the denominator ($\hat{Y}$); then
c) dividing the numerator estimate by the denominator estimate.

10.2.3 Tabulation of quantitative estimates

Estimates of sums or averages for quantitative variables can be obtained using the following three steps (only step a) is necessary to obtain the estimate of a sum):

a) multiplying the value of the variable of interest by the final weight and summing this quantity over all records of interest to obtain the numerator ($\hat{x}$);
b) summing the final weights of records having the characteristic of interest for the denominator ($\hat{Y}$); then
c) dividing the numerator estimate by the denominator estimate.

For example, to obtain the estimate of the average number of cigarettes smoked each day by individuals who smoke daily, first compute the numerator ($\hat{x}$) by summing the product between the value of variable **SMK_045** and the weight **WTS_M.** Next, sum this value over those records with a value of "daily" to the variable **SMK_005**. The denominator ($\hat{Y}$) is obtained by summing the final weight of those records with a value of "daily" to the variable **SMK_005**. Divide ($\hat{x}$) by ($\hat{Y}$) to obtain the average number of cigarettes smoked each day by daily smokers.

10.3 Guidelines for statistical analysis

The CCHS is based upon a complex design, with stratification and multiple stages of selection, and unequal probabilities of selection of respondents. Using data from such complex surveys presents problems to analysts because the survey design and the selection probabilities affect the estimation and variance calculation procedures that should be used. The use of the survey weights is the first step to ensuring that the proper results are obtained.

While many analysis procedures found in statistical packages allow weights to be used, the meaning or definition of the weight in these procedures can differ from what is appropriate in a sample survey framework. The end result is that, while in many cases the estimates produced by the packages are correct, the variances and the statistics based on those variances are almost meaningless. To get the proper results, the weights must be correctly interpreted by the software package and the proper variance estimation techniques must be used.

To ensure that results from this complex design are meaningful, there are several options. The first and most appropriate option is to use the variance calculation options outlined in Chapter 11 based on the bootstrap methodology. Second, survey specific procedures exist in many packages that properly interpret the weight. For example, in SAS, PROC SURVEYMEANS would provide results that are more appropriate as the bootstrap weights are used for the variance estimates than the results produced from PROC MEANS. PROC MEANS can use the survey weights to correctly adjust estimates, taking into account the unequal weighting design, but it tends to underestimate the variances in not using the bootstrap weights and assuming the design is simple random sampling. Please consult the bootstrap variance estimation document found in the CCHS release package for more information. It is recommended that data users use software packages and functions that can incorporate both sample weights and bootstrap weights while performing their analyses.

Only the options outlined in Chapter 11 based on the bootstrap methodology will take the stratification, clustering and multiple frame design into account when calculating the variance.

10.4 Release guidelines

Since the estimates obtained from the survey are based on a sample, there is variability in the values obtained in the sense that a different sample could result in different results. To take this into consideration, users should first ensure that there are enough observations to properly estimate the statistic and also to estimate the variance. Once the variance is obtained, users should ensure that the variance is reasonable enough that the estimate can properly be interpreted as being near the true population value.

Master or share file

For users of the master or share files, it is recommended to have at least 10 observations with the characteristic of interest and 20 in the domain if a proportion is being calculated. With enough observations, the user can proceed to calculating the variance and the coefficients of variation using the bootstrap weights provided with the data along with the appropriate software to do the

analysis. The variances can be used to calculate the CVs which aid in assessing the reliability of an estimate in regard to the sampling variability. The CV provides a relative measure of the sampling error as a proportion of the estimate. Estimates should be vetted using the guidelines in Table 10.1. CVs will increase as the variability of an estimate increases, and decrease as an estimate is more precise. However, an estimate can be precise while still having a large CV, specifically if the estimate has a value close to 0. Examining the confidence interval of the estimate will provide further indication of the quality of the estimate in terms of the variability. Long confidence intervals indicate less precision in the estimate while smaller confidence intervals indicate greater precision. When assessing the trustworthiness of sample proportions, the confidence intervals of estimates should be taken into account.

Table 10.1 Sampling variability guidelines

Type of Estimate	CV (in %)	Guidelines
Acceptable	CV $\leq$ 15.0	Estimates can be considered for general unrestricted release. Requires no special notation.
Marginal	15.0 < CV $\leq$ 35.0	Estimates can be considered for general unrestricted release but should be accompanied by a warning cautioning subsequent users of the high sampling variability associated with the estimates. Such estimates should be identified by the letter E (or in some other similar fashion).
Unacceptable	CV > 35.0	Statistics Canada recommends not to release estimates of unacceptable quality. However, if the user chooses to do so then estimates should be flagged with the letter F and the following warning should accompany the estimates: "*The user is advised that . . .(specify the data) . . . do not meet Statistics Canada's quality standards for this statistical program. Conclusions based on these data will be unreliable and most likely invalid. These data and any consequent findings should not be published. If the user chooses to publish these data or findings, then this disclaimer must be published with the data.*"

11. APPROXIMATE SAMPLING VARIABILITY TABLES

For a quick approximation of coefficients of variation that will be applicable to a wide variety of categorical estimates produced from the Share file, a set of Approximate Sampling Variability Tables will be produced. These "look-up" tables allow the user to obtain an approximate coefficient of variation based on the size of the estimate calculated from the survey data. Table 11.1 shows the CCHS 2017 look-up table at the Canada level as an example.

Table 11.1, CCHS Approximate Sampling Variability Tables, 2017 Share

NUMERATOR OF PERCENTAGE ('000)	ESTIMATED PERCENTAGE 0.1%	1.0%	2.0%	5.0%	10.0%	15.0%	20.0%	25.0%	30.0%	35.0%	40.0%	50.0%	70.0%	90.0%
1	126.8	126.2	125.5	123.6	120.3	116.9	113.4	109.8	106.1	102.2	98.2	89.7	69.5	40.1
2	89.6	89.2	88.8	87.4	85.1	82.7	80.2	77.7	75.0	72.3	69.5	63.4	49.1	28.4
3	73.2	72.9	72.5	71.4	69.5	67.5	65.5	63.4	61.3	59.0	56.7	51.8	40.1	23.2
4	63.4	63.1	62.8	61.8	60.2	58.5	56.7	54.9	53.1	51.1	49.1	44.8	34.7	20.1
5	56.7	56.4	56.1	55.3	53.8	52.3	50.7	49.1	47.5	45.7	43.9	40.1	31.1	17.9
6	51.7	51.5	51.3	50.5	49.1	47.7	46.3	44.8	43.3	41.7	40.1	36.6	28.4	16.4
7	47.9	47.7	47.5	46.7	45.5	44.2	42.9	41.5	40.1	38.6	37.1	33.9	26.3	15.2
8	44.8	44.6	44.4	43.7	42.5	41.3	40.1	38.8	37.5	36.1	34.7	31.7	24.6	14.2
9	42.3	42.1	41.8	41.2	40.1	39.0	37.8	36.6	35.4	34.1	32.7	29.9	23.2	13.4
10	40.1	39.9	39.7	39.1	38.0	37.0	35.9	34.7	33.6	32.3	31.1	28.4	22.0	12.7
11	38.2	38.0	37.9	37.3	36.3	35.3	34.2	33.1	32.0	30.8	29.6	27.0	20.9	12.1
12	36.6	36.4	36.2	35.7	34.7	33.8	32.7	31.7	30.6	29.5	28.4	25.9	20.1	11.6
13	35.2	35.0	34.8	34.3	33.4	32.4	31.5	30.5	29.4	28.4	27.2	24.9	19.3	11.1
14	33.9	33.7	33.6	33.0	32.2	31.2	30.3	29.4	28.4	27.3	26.3	24.0	18.6	10.7
15	32.7	32.6	32.4	31.9	31.1	30.2	29.3	28.4	27.4	26.4	25.4	23.2	17.9	10.4
16	31.7	31.5	31.4	30.9	30.1	29.2	28.4	27.5	26.5	25.6	24.6	22.4	17.4	10.0
17	30.7	30.6	30.4	30.0	29.2	28.4	27.5	26.6	25.7	24.8	23.8	21.7	16.8	9.7
18	29.9	29.7	29.6	29.1	28.4	27.6	26.7	25.9	25.0	24.1	23.2	21.1	16.4	9.5
19	29.1	28.9	28.8	28.4	27.6	26.8	26.0	25.2	24.3	23.5	22.5	20.6	15.9	9.2
20	28.3	28.2	28.1	27.6	26.9	26.1	25.4	24.6	23.7	22.9	22.0	20.1	15.5	9.0
21	27.7	27.5	27.4	27.0	26.3	25.5	24.8	24.0	23.2	22.3	21.4	19.6	15.2	8.8
22	27.0	26.9	26.8	26.4	25.7	24.9	24.2	23.4	22.6	21.8	20.9	19.1	14.8	8.6
23	26.4	26.3	26.2	25.8	25.1	24.4	23.7	22.9	22.1	21.3	20.5	18.7	14.5	8.4
24	25.9	25.8	25.6	25.2	24.6	23.9	23.2	22.4	21.7	20.9	20.1	18.3	14.2	8.2
25	25.4	25.2	25.1	24.7	24.1	23.4	22.7	22.0	21.2	20.4	19.6	17.9	13.9	8.0
30	23.1	23.0	22.9	22.6	22.0	21.3	20.7	20.1	19.4	18.7	17.9	16.4	12.7	7.3
35	21.4	21.3	21.2	20.9	20.3	19.8	19.2	18.6	17.9	17.3	16.6	15.2	11.7	6.8
40	20.0	20.0	19.9	19.5	19.0	18.5	17.9	17.4	16.8	16.2	15.5	14.2	11.0	6.3
45	18.9	18.8	18.7	18.4	17.9	17.4	16.9	16.4	15.8	15.2	14.6	13.4	10.4	6.0
50	17.9	17.8	17.8	17.5	17.0	16.5	16.0	15.5	15.0	14.5	13.9	12.7	9.8	5.7
55	17.1	17.0	16.9	16.7	16.2	15.8	15.3	14.8	14.3	13.8	13.2	12.1	9.4	5.4
60	16.4	16.3	16.2	16.0	15.5	15.1	14.6	14.2	13.7	13.2	12.7	11.6	9.0	5.2
65	15.7	15.7	15.6	15.3	14.9	14.5	14.1	13.6	13.2	12.7	12.2	11.1	8.6	5.0
70	15.2	15.1	15.0	14.8	14.4	14.0	13.6	13.1	12.7	12.2	11.7	10.7	8.3	4.8
75	14.6	14.6	14.5	14.3	13.9	13.5	13.1	12.7	12.3	11.8	11.3	10.4	8.0	4.6
80	14.2	14.1	14.0	13.8	13.5	13.1	12.7	12.3	11.9	11.4	11.0	10.0	7.8	4.5
85	13.7	13.7	13.6	13.4	13.0	12.7	12.3	11.9	11.5	11.1	10.7	9.7	7.5	4.3
90	13.4	13.3	13.2	13.0	12.7	12.3	12.0	11.6	11.2	10.8	10.4	9.5	7.3	4.2
95	13.0	12.9	12.9	12.7	12.3	12.0	11.6	11.3	10.9	10.5	10.1	9.2	7.1	4.1
100	12.7	12.6	12.6	12.4	12.0	11.7	11.3	11.0	10.6	10.2	9.8	9.0	6.9	4.0
125	11.3	11.3	11.2	11.1	10.8	10.5	10.1	9.8	9.5	9.1	8.8	8.0	6.2	3.6
150	10.3	10.3	10.3	10.1	9.8	9.5	9.3	9.0	8.7	8.3	8.0	7.3	5.7	3.3
200	9.0	8.9	8.9	8.7	8.5	8.3	8.0	7.8	7.5	7.2	6.9	6.3	4.9	2.8
250	8.0	8.0	7.9	7.8	7.6	7.4	7.2	6.9	6.7	6.5	6.2	5.7	4.4	2.5
300	7.3	7.3	7.2	7.1	6.9	6.8	6.5	6.3	6.1	5.9	5.7	5.2	4.0	2.3
350	********	6.7	6.7	6.6	6.4	6.2	6.1	5.9	5.7	5.5	5.3	4.8	3.7	2.1
400	********	6.3	6.3	6.2	6.0	5.8	5.7	5.5	5.3	5.1	4.9	4.5	3.5	2.0
450	********	5.9	5.9	5.8	5.7	5.5	5.3	5.2	5.0	4.8	4.6	4.2	3.3	1.9
500	********	5.6	5.6	5.5	5.4	5.2	5.1	4.9	4.7	4.6	4.4	4.0	3.1	1.8
750	**********		4.6	4.5	4.4	4.3	4.1	4.0	3.9	3.7	3.6	3.3	2.5	1.5
1000	**********		4.0	3.9	3.8	3.7	3.6	3.5	3.4	3.2	3.1	2.8	2.2	1.3
1500	**********		3.2	3.2	3.1	3.0	2.9	2.8	2.7	2.6	2.5	2.3	1.8	1.0
2000	**********			2.8	2.7	2.6	2.5	2.5	2.4	2.3	2.2	2.0	1.6	0.9
3000	**********			2.3	2.2	2.1	2.1	2.0	1.9	1.9	1.8	1.6	1.3	0.7
4000	**********				1.9	1.8	1.8	1.7	1.7	1.6	1.6	1.4	1.1	0.6
5000	**********					1.7	1.6	1.6	1.5	1.4	1.4	1.3	1.0	0.6
6000	**********					1.5	1.5	1.4	1.4	1.3	1.3	1.2	0.9	0.5
7000	**********						1.4	1.3	1.3	1.2	1.2	1.1	0.8	0.5
8000	**********							1.2	1.2	1.1	1.1	1.0	0.8	0.4
9000	**********							1.2	1.1	1.1	1.0	0.9	0.7	0.4
10000	**********								1.1	1.0	1.0	0.9	0.7	0.4
12500	**********										0.9	0.8	0.6	0.4
15000	**********										0.8	0.7	0.6	0.3

The coefficients of variation (CV) are derived using the variance formula for simple random sampling and incorporating a factor which reflects the multi-stage, clustered nature of the sample design. This factor, known as the ***design effect,*** was determined by first calculating design effects for a wide range of characteristics and then choosing, for each table produced, a conservative value among all design effects relative to that table. The value chosen was then used to generate a table that applies to the entire set of characteristics.

The Approximate Sampling Variability Tables, along with the design effects, the sample sizes and the population counts that were used to produce them, are provided in the document *Approximate*

Sampling Variability Tables, which is available to the share file users. All coefficients of variation in the Approximate Sampling Variability Tables are approximate and, therefore, unofficial. Options concerning the computation of exact coefficients of variation are discussed in sub-section 11.7. Analysis using bootstrap weights is the method that should be employed for more precise results.

Remember: As indicated in Sampling Variability Guidelines in Section 10.4, if the number of observations on which an estimate is based is less than 10, the weighted estimate should not be released regardless of the value of the coefficient of variation. Coefficients of variation based on small sample sizes are too unpredictable to be adequately represented. Using the tables to estimate coefficients of variation are even less precise than using the bootstrap weights, so in this case, a minimum of 30 observations is required.

11.1 How to use the CV tables for categorical estimates

The following rules should enable the user to determine the approximate coefficients of variation from the Sampling Variability Tables for estimates of the number, proportion or percentage of the surveyed population possessing a certain characteristic and for ratios and differences between such estimates.

Rule 1: Estimates of numbers possessing a characteristic (aggregates)
The coefficient of variation depends only on the size of the estimate itself. On the appropriate Approximate Coefficients of Variations Table, locate the estimated number in the left-most column of the table (headed "Numerator of Percentage") and follow the asterisks (if any) across to the first figure encountered. Since not all the possible values for the estimate are available, the smallest value which is the closest must be taken (as an example, if the estimate is equal to 1,700 and the two closest available values are 1,000 and 2,000, the first has to be chosen). This figure is the approximate coefficient of variation.

Rule 2: Estimates of proportions or percentages of people possessing a characteristic
The coefficient of variation of an estimated proportion (or percentage) depends on both the size of the proportion and the size of the numerator upon which the proportion is based. Estimated proportions are relatively more reliable than the corresponding estimates of the numerator of the proportion when the proportion is based upon a sub-group of the population. This is due to the fact that the coefficients of variation of the latter type of estimates are based on the largest entry in a row of a particular table, whereas the coefficients of variation of the former type of estimators are based on some entry (not necessarily the largest) in that same row. (Note that in the tables the CVs decline in value reading across a row from left to right). For example, the estimated proportion of individuals who smoke daily out of those who smoke at all is more reliable than the estimated number who smoke daily.

When the proportion (or percentage) is based upon the total population covered by each specific table, the CV of the proportion is the same as the CV of the numerator of the proportion. In this case, this is equivalent to applying Rule 1.

When the proportion (or percentage) is based upon a subset of the total population (e.g., those who smoke at all), reference should be made to the proportion (across the top of the table) and to the numerator of the proportion (down the left side of the table). Since not all the possible values for the proportion are available, the smallest value which is the closest must be taken (for example, if the proportion is 23% and the two closest values available in the column are 20% and 25%, 20% must be chosen). The intersection of the appropriate row and column gives the coefficient of variation.

Rule 3: Estimates of differences between aggregates or percentages

The standard error of a difference between two estimates is approximately equal to the square root of the sum of squares of each standard error considered separately. That is, the standard error of a difference ($\hat{d} = \hat{X}_2 - \hat{X}_1$) is:

$$\sigma_{\hat{d}} = \sqrt{(\hat{X}_1 \alpha_1)^2 + (\hat{X}_2 \alpha_2)^2}$$

where $\hat{X}_1$ is estimate 1, $\hat{X}_2$ is estimate 2, and α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively. The coefficient of variation of $\hat{d}$ is given by $\sigma_{\hat{d}} / \hat{d}$. This formula is accurate for the difference between independent populations or subgroups, but is only approximate otherwise. It will tend to overstate the error, if $\hat{X}_1$ and $\hat{X}_2$ are positively correlated and understate the error if $\hat{X}_1$ and $\hat{X}_2$ are negatively correlated.

Rule 4: Estimates of ratios

In the case where the numerator is a subset of the denominator, the ratio should be converted to a percentage and Rule 2 applied. This would apply, for example, to the case where the denominator is the number of individuals who smoke at all and the numerator is the number of individuals who smoke daily out of those who smoke at all.

Consider the case where the numerator is not a subset of the denominator, as for example, the ratio of the number of individuals who smoke daily or occasionally as compared to the number of individuals who do not smoke at all. The standard deviation of the ratio of the estimates is approximately equal to the square root of the sum of squares of each coefficient of variation considered separately multiplied by $\hat{R}$, where $\hat{R}$ is the ratio of the estimates ($\hat{R} = \hat{X}_1 / \hat{X}_2$). That is, the standard error of a ratio is:

$$\sigma_{\hat{R}} = \hat{R} \sqrt{\alpha_1^2 + \alpha_2^2}$$

Where α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively.

The coefficient of variation of $\hat{R}$ is given by $\sigma_{\hat{R}} / \hat{R} = \sqrt{\alpha_1^2 + \alpha_2^2}$. The formula will tend to overstate the error, if $\hat{X}_1$ and $\hat{X}_2$ are positively correlated and understate the error if $\hat{X}_1$ and $\hat{X}_2$ are negatively correlated.

Rule 5: Estimates of differences of ratios
In this case, Rules 3 and 4 are combined. The CVs for the two ratios are first determined using Rule 4, and then the CV of their difference is found using Rule 3.

11.2 Examples of using the CV tables for categorical estimates

The following "real life" examples are included to assist users in applying the foregoing rules. Unrounded numbers are used in the examples to facilitate the step by step instructions. These examples use the 2017 CCHS CV tables (Table 11.1). Users should still follow the rounding guidelines outlined in section 10.1 before publishing numbers.

Example 1: Estimates of numbers possessing a characteristic (aggregates)
Suppose that a user estimates that 3,665,449 individuals smoke daily in Canada. How does the user determine the coefficient of variation of this estimate?

1) Refer to the CANADA level CV table.
2) The estimated aggregate (3,665,449) does not appear in the left-hand column (the "Numerator of Percentage" column), so it is necessary to use the smallest figure closest to it, namely 3,000,000.
3) The coefficient of variation for an estimated aggregate (expressed as a percentage) is found by referring to the first non-asterisk entry on that row, namely, 2.3%.
4) So the approximate coefficient of variation of the estimate is 2.3%. According to the Sampling Variability Guidelines presented in Section 10.4, the finding that there were 3,665,449 individuals who smoke daily is publishable after applying the rounding rules.

Example 2: Estimates of proportions or percentages possessing a characteristic
Suppose that the user estimates that 3,665,449/5,151,237=71.2% of individuals in Canada who smoke at all smoke daily. How does the user determine the coefficient of variation of this estimate?

1) Refer to the CANADA level CV table.
2) Because the estimate is a percentage which is based on a subset of the total population (i.e., individuals who smoke at all, that is to say, daily or occasionally), it is necessary to use both the percentage (71.2%) and the numerator portion of the percentage (3,665,449) in determining the coefficient of variation.
3) The numerator (3,665,449) does not appear in the left-hand column (the "Numerator of Percentage" column) so it is necessary to use the smallest figure closest to it, namely 3,000,000. Similarly, the percentage estimate does not appear as any of the column headings, so it is necessary to use the smallest figure closest to it, 70.0%.
4) The figure at the intersection of the row and column used, namely 1.3% is the coefficient of variation (expressed as a percentage) to be used.
5) So the approximate coefficient of variation of the estimate is 1.3%. According to the Sampling Variability Guidelines presented in Section 10.4, the finding that 71.2% of individuals who smoke at all smoke daily can be published with no qualifications.

Example 3: Estimates of differences between aggregates or percentages

Suppose that a user estimates that, among men, 2,067955 /15,115,880 = 13.7% smoke daily (estimate 1), while for women, this percentage is estimated at 1,597,494 / 15,528,483 = 10.3% (estimate 2). How does the user determine the coefficient of variation of the difference between these two estimates?

1) Using the CANADA level CV table in the same manner as described in example 2 gives the CV for estimate 1 as 2.7% (expressed as a percentage), and the CV for estimate 2 as 3. 1% (expressed as a percentage).

2) Using rule 3, the standard error of a difference ($\hat{d} = \hat{X}_2 - \hat{X}_1$) is :

$$\sigma_{\hat{d}} = \sqrt{(\hat{X}_1 \alpha_1)^2 + (\hat{X}_2 \alpha_2)^2}$$

where $\hat{X}_1$ is estimate 1, $\hat{X}_2$ is estimate 2, and α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively. The standard error of the difference $\hat{d}$ = (0.137 - 0.103) = 0.034 is:

$$\sigma_{\hat{d}} = \sqrt{[(0.137)(0.027)]^2 + [(0.103)(0.031)]^2}$$

= 0.00489

3) The coefficient of variation of $\hat{d}$ is given by $\sigma_{\hat{d}} / \hat{d}$ = 0.00489/0.034 = 0.144.
4) So the approximate coefficient of variation of the difference between the estimates is 14.6% (expressed as a percentage). According to the Sampling Variability Guidelines presented in Section 10.4, this estimate can be published.

Example 4: Estimates of ratios

Suppose that the user estimates that 3,665,449 individuals smoke daily, while 1,485,788 individuals smoke occasionally. The user is interested in comparing the estimate of daily to occasional smokers in the form of a ratio. How does the user determine the coefficient of variation of this estimate?

1) First of all, this estimate is a ratio estimate, where the numerator of the estimate (= $\hat{X}_1$) is the number of individuals who smoke occasionally. The denominator of the estimate (= $\hat{X}_2$) is the number of individuals who smoke daily.
2) Refer to the CANADA level CV table.
3) The numerator of this ratio estimate is 1,485,788. The smallest figure closest to it is 1,000,000. The coefficient of variation for this estimate (expressed as a percentage) is found by referring to the first non-asterisk entry on that row, namely, 4.0%.
4) The denominator of this ratio estimate is 3,665,449. The figure closest to it is 3,000,000. The coefficient of variation for this estimate (expressed as a percentage) is found by referring to the first non-asterisk entry on that row, namely, 2.3%.
5) So the approximate coefficient of variation of the ratio estimate is given by rule 4:

$$\alpha_{\hat{R}} = \sqrt{\alpha_1^{\ 2} + \alpha_2^{\ 2}},$$

That is,

$$\alpha_{\hat{R}} = \sqrt{(0.040)^2 + (0.023)^2}$$

= 0.046

where α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively. The obtained ratio of occasional to daily smokers is 1,485,788/3,665,449 which is 0.41:1. The coefficient of variation of this estimate is 4.6% (expressed as a percentage), which is releasable with no qualifications, according to the Sampling Variability Guidelines presented in Section 10.4.

11.3 How to use the CV tables to obtain confidence limits

Although coefficients of variation are widely used, a more intuitively meaningful measure of sampling error is the confidence interval of an estimate. A confidence interval constitutes a statement on the level of confidence that the true value for the population lies within a specified range of values. For example a 95% confidence interval can be described as follows: if sampling of the population is repeated indefinitely, each sample leading to a new confidence interval for an estimate, then in 95% of the samples the interval will cover the true population value.

Using the standard error of an estimate, confidence intervals for estimates may be obtained under the assumption that under repeated sampling of the population, the various estimates obtained for a population characteristic are normally distributed about the true population value. Under this assumption, the chances are about 68 out of 100 that the difference between a sample estimate and the true population value would be less than one standard error, about 95 out of 100 that the difference would be less than two standard errors, and about 99 out of 100 that the differences would be less than three standard errors. These different degrees of confidence are referred to as the confidence levels.

Confidence intervals for an estimate, $\hat{X}$, are generally expressed as two numbers, one below the estimate and one above the estimate, as $(\hat{X} - k, \hat{X} + k)$, where k is determined depending upon the level of confidence desired and the sampling error of the estimate.

Confidence intervals for an estimate can be calculated directly from the Approximate Sampling Variability Tables by first determining from the appropriate table the coefficient of variation of the estimate $\hat{X}$, and then using the following formula to convert to a confidence interval CI:

$$CI_X = [\ \hat{X} - z\ \hat{X}\ \alpha_{\hat{X}}, \hat{X} + z\ \hat{X}\ \alpha_{\hat{X}}\ ]$$

Where $\alpha_{\hat{X}}$ is determined coefficient of variation for $\hat{X}$, and

z = 1 if a 68% confidence interval is desired
z = 1.6 if a 90% confidence interval is desired
z = 2 if a 95% confidence interval is desired
z = 3 if a 99% confidence interval is desired.

Note: Release guidelines presented in section 10.4 which apply to the estimate also apply to the confidence interval. For example, if the estimate is not releasable, then the confidence interval is not releasable either.

11.4 Example of using the CV tables to obtain confidence limits

A 95% confidence interval for the estimated proportion of individuals who smoke daily from those who smoke at all (from example 2, sub-section 11.2) would be calculated as follows:

$\hat{X}$ = 0.712

z = 2

$\alpha_{\hat{X}}$ = 0.013 is the coefficient of variation of this estimate as determined from the tables.

$CI_{\hat{x}}$ = {0.712 - (2) (0.712) (0.013), 0.712 + (2) (0.712) (0.013)}

$CI_{\hat{x}}$ = {0.693, 0.731}

11.5 How to use the CV tables to do a Z-test

Standard errors may also be used to perform hypothesis testing, a procedure for distinguishing between population parameters using sample estimates. The sample estimates can be numbers, averages, percentages, ratios, etc. Tests may be performed at various levels of significance, where a level of significance is the probability of concluding that the characteristics are different when, in fact, they are identical.

Let $\hat{X}_1$ and $\hat{X}_2$ be sample estimates for two characteristics of interest. Let the standard error on the difference $\hat{X}_1 - \hat{X}_2$ be $\sigma_{\hat{d}}$. If the ratio of $\hat{X}_1 - \hat{X}_2$ over $\sigma_{\hat{d}}$ is between -2 and 2, then no conclusion about the difference between the characteristics is justified at the 5% level of significance. If however, this ratio is smaller than -2 or larger than +2, the observed difference is significant at the 5% level.

11.6 Example of using the CV tables to do a Z-test

Let us suppose we wish to test, at 5% level of significance, the hypothesis that there is no difference between the proportion of men who smoke daily AND the proportion of women who smoke daily. From example 3, sub-section 11.2, the standard error of the difference between these two estimates was found to be = 0.00489. Hence,

$$z = \frac{\hat{X}_1 - \hat{X}_2}{\sigma_{\hat{d}}} = \frac{0.137 - 0.103}{0.0048995} = \frac{0.034}{0.0048995} = 6.87$$

Since z = 6.95 is greater than 2, it must be concluded that there is a significant difference between the two estimates at the 5% level of significance. Note that the two sub-groups compared are considered as being independent, so the test is valid.

11.7 Exact variances/coefficients of variation

All coefficients of variation in the Approximate Sampling Variability Tables (CV Tables) are indeed approximate and, therefore, unofficial.

The computation of exact coefficients of variation is not a straightforward task since there is no simple mathematical formula that would account for all CCHS sampling frame and weighting aspects. Therefore, other methods such as resampling methods must be used in order to estimate measures of precision. Among these methods, the bootstrap method is the one recommended for analysis of CCHS data.

Many statistical packages allow for analyses using the bootstrap weights. More details are provided in section 10.3 and in the document on bootstrap variance estimation.

There are a number of reasons why a user may require an exact variance. A few are given below. Firstly, if a user desires estimates at a geographic level other than those available in the tables (for example, at the rural/urban level), then the CV tables provided are not adequate. Coefficients of variation of these estimates may be obtained using "domain" estimation techniques through the exact variance program.

Secondly, should a user require more sophisticated analyses such as estimates of parameters from linear regressions or logistic regressions, the CV tables will not provide correct associated coefficients of variation. Although some standard statistical packages allow sampling weights to be incorporated in the analyses, the variances that are produced often do not take into account the stratified and clustered nature of the design properly, whereas the exact variance program would do so.

Thirdly, for estimates of quantitative variables, separate tables are required to determine their sampling error. Since most of the variables for the CCHS are primarily categorical in nature, this has not been done. Thus, users wishing to obtain coefficients of variation for quantitative variables can do so through the exact variance program. As a general rule, however, the coefficient of variation of a quantitative total will be larger than the coefficient of variation of the corresponding category estimate (i.e., the estimate of the number of persons contributing to the quantitative estimate). If the corresponding category estimate is not releasable, the quantitative estimate will not be either. For example, the coefficient of variation of the estimate of the total number of cigarettes smoked each day by individuals who smoke daily would be greater than the coefficient of variation of the corresponding estimate of the number of individuals who smoke daily. Hence if the coefficient of variation of the latter is not releasable, then the coefficient of variation of the corresponding quantitative estimate will also not be releasable.

Coefficients of variation produced by the tables are based on a wide range of variables and are therefore considered crude, whereas the exact variance program would give an exact coefficient of variation associated with the variable in question.

12. MICRODATA FILES: DESCRIPTION, ACCESS AND USE

The CCHS produces three types of microdata files: master files, share files and public use microdata files (PUMF). Table 12.1 includes the list of all available 2021 collection period data files.

12.1 Master files

The master files contain all variables and all records from the survey for an associated time frame. These files are accessible at Statistics Canada for internal use and in Statistics Canada's Research Data Centres (RDC), and are also subject to custom tabulation requests.

12.1.1 Research Data Centre

The RDC Program enables researchers to use the survey data in the master files in a secure environment in several universities across Canada. Researchers must submit research proposals that, once approved, give them access to the RDC. For more information, please consult the following web page: https://www.statcan.gc.ca/eng/rdc/index

12.1.2 Custom tabulations

Another way to access the master files is to offer all users the option of having staff in Client Services of the Centre for Population Health Data prepare custom tabulations. This service is offered on a cost-recovery basis. It allows users who do not possess knowledge of tabulation software products to get custom results. The results are screened for confidentiality and reliability concerns before release. For more information, please contact Client Services at 613-951-1746 or by e-mail at statcan.hd-ds.statcan@canada.ca.

12.1.3 Remote access

The remote access service to the survey master files is another way to have access to these data if, for some reason, the user cannot access a Research Data Centre but have an approved research proposal. A researcher can be supplied with a synthetic or 'dummy' master file and a corresponding record layout. With these tools, the researcher can develop his or her own set of analytical computer programs. The code for the custom tabulations is then sent via e-mail to statcan.hd-ds.statcan@canada.ca. The code will then be transferred into Statistics Canada's internal secured network and processed using the appropriate master file of CCHS data. Estimates generated will be released to the user, subject to meeting the guidelines for analysis and release outlined in Section 10 of this document. Results are screened for confidentiality and reliability concerns and then the output is returned to the client. There is no charge for this service.

12.1.4 Real Time Remote Access (RTRA)

Users can access CCHS data using the Real Time Remote Access (RTRA) system, enabling fast access to Statistics Canada microdata. The RTRA system is an on-line remote access facility allowing users to run SAS programs, in real-time, against micro-data sets located in a central and secure location. Researchers using the RTRA system do not gain direct access to the micro-data

and cannot view the content of the micro-data file. Instead, users submit SAS programs to extract results in the form of frequency tables. As RTRA researchers cannot view the micro-data, becoming a deemed employee of Statistics Canada is no longer necessary. This relationship is the basis that allows the RTRA to service its clients rapidly. Please see https://www.statcan.gc.ca/eng/rtra/rtra for more information.

12.2 Share files

The share files contain all variables and all records of CCHS respondents who agreed to share their data with Statistic Canada's partners, which are Health Canada and the Public Health Agency of Canada, as well as the provincial and territorial ministries of health. The share file is released only to these organizations. Personal identifiers are removed from the share files to respect respondent confidentiality. Users of these files must first certify that they will not disclose, at any time, any information that might identify a survey respondent.

12.3 Public use microdata files

The public use microdata files (PUMF) are developed from the master files using a technique that balances the need to ensure respondent confidentiality with the need to produce the most useful data possible at the health region level. The PUMF must meet stringent security and confidentiality standards required by the *Statistics Act* before they are released for public access. To ensure that these standards have been achieved, each PUMF goes through a formal review and approval process by an executive committee of Statistics Canada.

Variables most likely to lead to identification of an individual are deleted from the data file or are collapsed to broader categories. Due to the risk of disclosure household weights are not included on the PUMF.

The PUMF contains the data collected over two years. It includes questions that were asked over two years. Unless otherwise specified, these questions are usually those included in the annual core content and in the two-year theme content as well as the optional content selected for two years by the provinces and territories.

There is no charge to access the PUMF in a post-secondary educational institution that is part of the Data Liberation Initiative, a partnership between post-secondary institutions and Statistics Canada for improving access to Canadian data resources. For more information on the Data Liberation Initiative please see http://www.statcan.gc.ca/eng/dli/dli.

The CCHS PUMF files are also free of charge to any researcher that contacts Client Services at 613-951-1746 or by e-mail at statcan.hd-ds.statcan@canada.ca.

Table 12.1 2021 CCHS data files

Reference Period	Files	File Type	File name	Sampling weight	Bootstrap weights file	Variables included	Records included
2021	Main file	Master	HS.txt	WTS_M	bsw.txt	All core and all optional / theme **m**odules	All respondent records in the ten provinces
		Share		WTS_S			Records of all respondents who agreed to share their data
January to April 2021	Rapid response file (SCH2/TAV)	Master	HS_SCH2.txt	WTS_M	bsw.txt	All core modules and questions in the SCH2 & TAV rapid response	All respondent records in the ten provinces
		Share		WTS_S			Records of all respondents who agreed to share their data

12.4 How to use the CCHS data files: annual data file or two-year data file?

Users who have access to share files or master files have the choice of using one-year or two-year data files. Decisions about which period to use in a given data analysis should be guided by the level of detail and the quality required. With a one-year file, estimates will not always be available because of the quality associated with limited sample sizes. Since the territories are not available on the one-year file, users wanting true national estimates should use the two-year file.
Before interpreting and using a CCHS estimate, it is recommended to make sure that the estimates meets the following rules:

- Coefficient of Variation 35.0% or less
- a minimum of 10 respondents in the domain with the characteristic and
- total domain of interest includes at least 20 respondents (for proportions or ratios)

This will not be possible for rare characteristics and detailed domains with one-year files. Instead, users will have to rely on two-year files or multi-year files. For more information on combining CCHS cycles into a multi-year file, please see https://www150.statcan.gc.ca/n1/pub/82-003-x/2009001/article/10795-eng.pdf

Where the use of either a one-year or two-year file is viable, the user should consider the trade-off between accuracy and timeliness. If it is important to reflect the current characteristics of a population as closely as possible, the one-year file would be preferable. However, with the increased sample size, more detailed estimates and analyses can be carried out with a two-year file.

12.5 Use of weight variable

The weight variable **WTS_M** represents the sampling weight for key survey files. For a given respondent, the sampling weight can be interpreted as the number of people the respondent represents in the Canadian population. This weight must always be used when computing statistical estimates in order to make inference at the population level possible. The production of

unweighted estimates is not recommended. The sample allocation, as well as the survey design specifics can cause such results to not correctly represent the population. Refer to section 8 on weighting for a more detailed explanation on the creation of this weight. The weight variable **WTS_M** must be used for regional analyses.

The Food security (FSC) module, included in certain reference period data files, measures concepts that apply not only to the respondent's situation, but also to that of the respondent's entire household. Depending on the level of analysis, the analysis of the variables may require use of a weight calculated to represent the number of Canadian households, rather than the number of persons. This weight variable **WTS_MHH** is found in a separate file (HS_HHWT.txt). It can be used in place of the variable **WTS_M** for household analyses at the national and provincial levels.

12.6 Variable naming convention

The variable naming convention adopted allows data users to easily use and identify the data based on the module and variable type. The CCHS variable naming convention fulfils two requirements: to restrict variable names to a maximum of eight characters for ease of use by analytical software products and to identify easily conceptually identical variables from one survey collection period to the next. Questions to which changes are made between two collection periods, and where the changes alter the concept measured by the question, are entirely renamed to avoid any confusion in the analysis.

A variable (REFPER, format = YYYYMM-YYYYMM) was added to the microdata files in order to identify the beginning and the end of the reference during which data included in the file were collected. This variable will be useful, notably for users wanting to use data from several collection periods at a time. Therefore, variable names for identical modules or questions from one collection year to the next (e.g., 2015 and 2017) will be the same. However, some minor changes have been made to variable names since the 2015 redesign.

The naming convention used for variables beginning with the 2015 CCHS use up to eight characters. The variable names are structured as follows:

Positions 1 to 3 :	Module/questionnaire section name
Position 4 :	Variable type (underscore, C,D, F or G)
Positions 5 to 8 :	Question number and answer option for multiple response questions in increments of five (e.g. 005, 010, 015)

Example 1 shows that the structure of the variable name for question 005, Smoking Module, is SMK_005 :

Positions 1 to 3 :	SMK	Smoking module
Position 4 :	_	(underscore = collected data)
Position 5 to 8 :	005	Question number

Example 2 shows the structure of the variable name for question 2 of the Contacts with health professionals – part 1 Module (CHP_015), which is a multi-response question:

Positions 1 to 3 :	CHP	Health care utilization module
Position 4 :	_	(underscore = collected data)
Position 5 to 8 :	015	Corresponding question number and answer option

Positions 1 to 3 contain the acronyms for each of the modules. These acronyms appear beside the module names given in the table in Appendix A. It should be noted that some module acronyms consist of four characters. Variables names from these modules are still bound by a maximum of 8 characters.
Position 4 designates the variable type based on whether it is a variable collected directly from a questionnaire question ("_"), derived ("D"), flag ("F"), or grouped ("G") variable.
In general, the last four positions (5 to 8) follow the variable numbering used on the questionnaire.

The letter "Q" used to represent the word "question" is removed, and all question numbers are presented in a two or three digit format. For example, question Q005A in the questionnaire becomes simply 005A, and question Q010 becomes simply 010.

Table 12.2 Designation of codes used in the 4th position of the CCHS variable names

_	Collected variable	A variable that appears directly on the questionnaire
C	Coded variable	A variable coded from one or more collected variables (e.g., SIC, Standard Industrial Classification code)
D	Derived variable	A variable calculated from one or more collected or coded variables, usually calculated during head office processing (e.g., Health Utility Index)
F	Flag variable	A variable calculated from one or more collected variables (like a derived variable), but usually calculated by the data collection computer application for later use during the interview (e.g., work flag)
G	Grouped variable	Collected, coded, suppressed or derived variables collapsed into groups (e.g., age groups)

For questions that have more than one response option, the final position in the variable naming sequence is represented by a letter. For this type of question, new variables were created to differentiate between a "yes" or "no" answer for each response option. For example, if Q010 had 4 response options, the new questions would be named Q010A for option 1, Q010B for option 2, Q010C for option 3, and Q010D for option 4. If only options 2 and 3 were selected, then Q010A = No, Q010B = Yes, Q010C = Yes and Q010D = No.

12.7 Data dictionaries

Separate data dictionary reports, including variable names, concepts, universe statements and frequencies, are provided for the main master and share files and each of the rapid response files (if applicable). All variables on these files will appear in the data dictionary report, though some may not include a frequency (SampleID for example).

When a variable includes the frequencies for responses, the report will show four columns:

1. Content – The text label for each of the response categories.
2. Code – The numerical value associated with each category.
3. Sample – The unweighted frequency on the file of respondents with each response.
4. Population – The weighted frequency for the response.

For each module in the data dictionary reports, a flag is used to indicate the availability of the module for each respondent. When the flag is equal to 2 (No), all variables in the module have "not applicable" values. For example, the DOWST variable indicates if the Work stress module applies to a given respondent.

12.8 Differences in calculation of core and theme content variables using different files

Variables from core content modules can be estimated using either of the two data files provided, when a one year and a two-year data file is available. Depending on which file is used, very small differences will be observed.

All official Statistics Canada estimates of variables from common modules are based on the main master file sampling weight.

APPENDIX A - CANADIAN COMMUNITY HEALTH SURVEY (CCHS) - CONTENT OVERVIEW - 2021

Content availability by province and territory – 2021

		2021												
Acronym	Module Name	NL (10)	PEI (11)	NS (12)	NB (13)	QC (24)	ON (35)	MB (46)	SK (47)	AB (48)	BC (59)	YK[1] (60)	NWT[1] (61)	NU[1] (62)
ADL	Activities of daily living	T	T	T	T	T	T	T	T	T	T	T	T	T
ADM	Administration information	C	C	C	C	C	C	C	C	C	C	C	C	C
HN	Health number validation	C	C	C	C	C	C	C	C	C	C	C	C	C
INC	Income	C	C	C	C	C	C	C	C	C	C	C	C	C
THI	Total household income	C	C	C	C	C	C	C	C	C	C	C	C	C
ALC	Alcohol use	C	C	C	C	C	C	C	C	C	C	C	C	C
ANC1	Age of respondent	C	C	C	C	C	C	C	C	C	C	C	C	C
ANC	Age without confirmation	C	C	C	C	C	C	C	C	C	C	C	C	C
DATE	Date	C	C	C	C	C	C	C	C	C	C	C	C	C
UN	Update name	C	C	C	C	C	C	C	C	C	C	C	C	C
CAN	Cannabis use	C	C	C	C	C	C	C	C	C	C	C	C	C
CCC	Chronic conditions	C/T	C/T	C/T	C/T	C/T	C/T	C/T	C/T	C/T	C/T	C/T	C/T	C/T
COV[2]	COVID-19 (202101-202108)	T	T	T	T	T	T	T	T	T	T	T	T	T
COV2[2]	COVID-19 (202109-202112)	T	T	T	T	T	T	T	T	T	T	T	T	T
PVC[3]	Vaccination passeport COVID-19 (202109-202112)	T	T	T	T	T	T	T	T	T	T	T	T	T
COV3[3]	COVID-19 (202109-202112)	T	T	T	T	T	T	T	T	T	T	T	T	T
DEP	Depression	O	O	O	O	O	O	O	O	O	O	O	O	O
DRG	Drug use	O	O	O	O	O	O	O	O	O	O	O	O	O
ECV	E-cigarettes and vaping	C	C	C	C	C	C	C	C	C	C	C	C	C
FSC	Food security	T	T	T	T	T	T	T	T	T	T	T	T	T
FVC	Fruit and vegetable consumption	T	T	T	T	T	T	T	T	T	T	T	T	T
GEN	General health	C	C	C	C	C	C	C	C	C	C	C	C	C
HM	Health minimum block	C	C	C	C	C	C	C	C	C	C	C	C	C
SLM	Subjective well-being minimum	C	C	C	C	C	C	C	C	C	C	C	C	C
GDR	Sex and Gender	C	C	C	C	C	C	C	C	C	C	C	C	C
GR	Proxy interview	C	C	C	C	C	C	C	C	C	C	C	C	C
HMC	Home care services - with palliative care	T	T	T	T	T	T	T	T	T	T	T	T	T
HWT	Height and weight - self-reported	C	C	C	C	C	C	C	C	C	C	C	C	C
LBF	Labour force	C	C	C	C	C	C	C	C	C	C	C	C	C
LMAM	Labour market activities minimal	C	C	C	C	C	C	C	C	C	C	C	C	C
LMA3	Class of worker	C	C	C	C	C	C	C	C	C	C	C	C	C
LMA4	Industry	C	C	C	C	C	C	C	C	C	C	C	C	C
LMA5	Occupation	C	C	C	C	C	C	C	C	C	C	C	C	C
LMA6	Usual hours of work	C	C	C	C	C	C	C	C	C	C	C	C	C
LM[3]	Telework (202106-202112)	T	T	T	T	T	T	T	T	T	T	T	T	T
MAC	Main activity	C	C	C	C	C	C	C	C	C	C	C	C	C

Acronym	Module Name	2021												
		NL (10)	PEI (11)	NS (12)	NB (13)	QC (24)	ON (35)	MB (46)	SK (47)	AB (48)	BC (59)	YK[1] (60)	NWT[1] (61)	NU[1] (62)
ESC1	Education - school attendance v.1	C	C	C	C	C	C	C	C	C	C	C	C	C
MED	Medication use	T	T	T	T	T	T	T	T	T	T	T	T	T
MEX	Maternal experiences	T	T	T	T	T	T	T	T	T	T	T	T	T
MXA	Alcohol use during maternal experience	T	T	T	T	T	T	T	T	T	T	T	T	T
MXS	Smoking during maternal experience	T	T	T	T	T	T	T	T	T	T	T	T	T
PAA	Physical activities - adults 18 years and older	T	T	T	T	T	T	T	T	T	T	T	T	T
PAY	Physical activities for youth	T	T	T	T	T	T	T	T	T	T	T	T	T
PAD	Physical activity detail (PAY sub-block)	T	T	T	T	T	T	T	T	T	T	T	T	T
PHC	Primary health care	O	O	O	O	O	O	O	O	O	O	O	O	O
PMK	Person most knowledgeable about household situation	C	C	C	C	C	C	C	C	C	C	C	C	C
REL	Relationship matrix	C	C	C	C	C	C	C	C	C	C	C	C	C
RNC	Relationship without confirmation	C	C	C	C	C	C	C	C	C	C	C	C	C
SBE	Sedentary behaviours	T	T	T	T	T	T	T	T	T	T	T	T	T
SCH2	Smoking - Stages of change 2	T	T	T	T	T	T	T	T	T	T	T	T	T
SDC	Socio-demographic characteristics	C	C	C	C	C	C	C	C	C	C	C	C	C
IMM	Immigration minimum block	C	C	C	C	C	C	C	C	C	C	C	C	C
AMB	Aboriginal minimum	C	C	C	C	C	C	C	C	C	C	C	C	C
PG	Population group	C	C	C	C	C	C	C	C	C	C	C	C	C
LAN	Language minimum	C	C	C	C	C	C	C	C	C	C	C	C	C
LLU	Language look-up	C	C	C	C	C	C	C	C	C	C	C	C	C
SLP	Sleep	T	T	T	T	T	T	T	T	T	T	T	T	T
SMK	Smoking	C	C	C	C	C	C	C	C	C	C	C	C	C
SPS	Social provisions (reduced)	O	O	O	O	O	O	O	O	O	O	O	O	O
TAL2	Tobacco products alternatives 2	T	T	T	T	T	T	T	T	T	T	T	T	T

Legend - Content Type	
C	Core
T	Theme
O	Optional
(blank)	Optional - Not selected

[1] Although they are collected every year, data of respondents living in the territories are only included on two year microdata files.

[2] The COV2 module, introduced in the September collection period, included updates to account for more detailed questions related to number of doses received and reasons for not getting vaccinated. The annual 2021 file only keeps questions asked consistently for the whole year (ie. same concept in COV and COV2). Variable names use the COV2 module acronym, regardless of collection period.

[3] These modules were not asked all year, therefore the data is not available on CCHS 2021 files.

APPENDIX B - AVAILABLE GEOGRAPHY IN THE MASTER AND SHARE FILES AND THEIR CORRESPONDING CODES: CANADA, PROVINCES, HEALTH REGIONS AND PEER GROUPS

Appendix B - Available geography in the master and share files and their corresponding codes: Canada, provinces, health regions and peer groups

0	Canada	
10	Newfoundland and Labrador	
1011-C		Eastern Regional Integrated Health Authority
1012-E		Central Regional Integrated Health Authority
1013-E		Western Regional Integrated Health Authority
1014-E		Labrador-Grenfell Regional Integrated Health Authority
11	Prince Edward Island	
1100-C		Prince Edward Island
12	Nova Scotia	
1201-E		Zone 1 - Western
1202-C		Zone 2 - Northern
1203-E		Zone 3 - Eastern
1204-A		Zone 4 - Central
13	New Brunswick	
1301-C		Zone 1 (Moncton area)
1302-A		Zone 2 (Saint John area)
1303-C		Zone 3 (Fredericton area)
1304-E		Zone 4 (Edmundston area)
1305-E		Zone 5 (Campbellton area)
1306-E		Zone 6 (Bathurst area)
1307-E		Zone 7 (Miramichi area)
24	Quebec	
2401-C		Région du Bas-Saint–Laurent
2402-C		Région du Saguenay – Lac–Saint–Jean
2403-A		Région de la Capitale-Nationale
2404-C		Région de la Mauricie et du Centre-du-Québec
2405-C		Région de l'Estrie
2406-G		Région de Montréal
2407-B		Région de l'Outaouais
2408-C		Région de l'Abitibi-Témiscamingue
2409-C		Région de la Côte-Nord
2410-C		Région du Nord-du-Québec
2411-E		Région de la Gaspésie — Îles-de-la-Madeleine
2412-D		Région de la Chaudière–Appalaches
2413-A		Région de Laval
2414-B		Région de Lanaudière
2415-B		Région des Laurentides
2416-B		Région de la Montérégie
35	Ontario by Local Health Integration Network	
3501		Erie St. Clair
3502		South West
3503		Waterloo Wellington
3504		Hamilton Niagara Haldimand Brant

3505		Central West
3506		Mississauga Halton
3507		Toronto Central
3508		Central
3509		Central East
3510		South East
3511		Champlain
3512		North Simcoe Muskoka
3513		North East
3514		North West
35	Ontario by Health Units	
3526-E		The District of Algoma Health Unit
3527-A		Brant County Health Unit
3530-B		Durham Regional Health Unit
3533-D		Grey Bruce Health Unit
3534-D		Haldimand-Norfolk Health Unit
3535-D		Haliburton, Kawartha, Pine Ridge District Health Unit
3536-B		Halton Regional Health Unit
3537-A		City of Hamilton Health Unit
3538-C		Hastings and Prince Edward Counties Health Unit
3539-D		Huron County Health Unit
3540-E		Chatham-Kent Health Unit
3541-C		Kingston, Frontenac and Lennox and Addington Health Unit
3542-C		Lambton Health Unit
3543-D		Leeds, Grenville and Lanark District Health Unit
3544-A		Middlesex–London Health Unit
3546-A		Niagara Regional Area Health Unit
3547-C		North Bay Parry Sound District Health Unit
3549-C		Northwestern Health Unit
3551-B		City of Ottawa Health Unit
3553-H		Peel Regional Health Unit
3554-D		Perth District Health Unit
3555-C		Peterborough County–City Health Unit
3556-C		Porcupine Health Unit
3557-D		Renfrew County and District Health Unit
3558-D		The Eastern Ontario Health Unit
3560-B		Simcoe Muskoka District Health Unit
3561-C		Sudbury and District Health Unit
3562-C		Thunder Bay District Health Unit
3563-C		Timiskaming Health Unit
3565-B		Waterloo Health Unit
3566-B		Wellington–Dufferin-Guelph Health Unit
3568-A		Windsor–Essex County Health Unit
3570-H		York Regional Health Unit
3575-D		Oxford Elgin St. Thomas Health Unit

3595-G		City of Toronto Health Unit
46	Manitoba	
4601-A		Winnipeg Regional Health Authority
4602-D		Prairie Mountain Health
4603-D		Interlake-Eastern Regional Health
4604-F		Northern Regional Health Authority
4605-B		Southern Health
47	Saskatchewan	
4701-D		Sun Country Regional Health Authority
4702-D		Five Hills Regional Health Authority
4703-D		Cypress Regional Health Authority
4704-B		Regina Qu'Appelle Regional Health Authority
4705-D		Sunrise Regional Health Authority
4706-B		Saskatoon Regional Health Authority
4707-D		Heartland Regional Health Authority
4708-C		Kelsey Trail Regional Health Authority
4709-C		Prince Albert Parkland Regional Health Authority
4710-B		Prairie North Regional Health Authority
4714-F		Mamawetan/Keewatin/Athabasca Regional Health Authorities[8]
48	Alberta	
4831-B		South Zone
4832-B		Calgary Zone
4833-B		Central Zone
4834-B		Edmonton Zone
4835-I		North Zone
59	British Columbia	
5911-D		East Kootenay Health Service Delivery Area
5912-C		Kootenay-Boundary Health Service Delivery Area
5913-A		Okanagan Health Service Delivery Area
5914-C		Thompson/Cariboo Health Service Delivery Area
5921-B		Fraser East Health Service Delivery Area
5922-H		Fraser North Health Service Delivery Area
5923-H		Fraser South Health Service Delivery Area
5931-H		Richmond Health Service Delivery Area
5932-G		Vancouver Health Service Delivery Area
5933-A		North Shore/Coast Garibaldi Health Service Delivery Area
5941-A		South Vancouver Island Health Service Delivery Area
5942-C		Central Vancouver Island Health Service Delivery Area

[8] For most data sources (with the exception of Census and Demographic population estimates), health region level data are not available for some northern health regions in Saskatchewan which have small populations. To avoid suppression in these areas where small numbers or sample size impact on data quality, data have been grouped with neighbouring regions, as follows: Athabasca Health Authority, Saskatchewan (4713) is combined with Mamawetan Churchill River Regional Health Authority (4711) and Keewatin Yatthé Regional Health Authority (4712) and referred to as "Mamawetan/Keewatin/Athabasca Regional Health Authorities (4714)"

5943-C		North Vancouver Island Health Service Delivery Area
5951-C		Northwest Health Service Delivery Area
5952-C		Northern Interior Health Service Delivery Area
5953-I		Northeast Health Service Delivery Area
A	Peer group A	
B	Peer group B	
C	Peer group C	
D	Peer group D	
E	Peer group E	
F	Peer group F	
G	Peer group G	
H	Peer group H	
I	Peer group I	

APPENDIX C - SAMPLE ALLOCATION BY PROVINCE/TERRITORY (CCHS 2021)

Appendix C – Sample allocation by province/territory (CCHS 2021)[9]

Province/Territory	CCHS		Total
	Adults	Youths	
Canada	200,847	12,931	213,778
N.L.	13,525	439	13,964
P.E.I.	5,435	253	5,688
N.S.	10,922	452	11,374
N.B.	11,376	440	11,816
Que.	25,284	2,142	27,426
Ont.	53,471	4,154	57,625
Man.	16,474	683	17,157
Sask.	17,691	628	18,319
Alta.	22,384	1,568	23,952
B.C.	22,675	1,925	24,600
Y.T.	706	103	809
N.W.T.	640	88	728
Nvt.	264	56	320

[9] These raw sample sizes correspond to the number of units that were sent to collection.

APPENDIX D - RESPONSE RATES BY PROVINCE/TERRITORY (CCHS 2021)

Appendix D – Response rates by province/territory (CCHS 2021)

Province/ Territory	Adults			Youth			Combined
	# in-scope	# respondents	Response rates	# in scope	# respondents	Response rates	Response rates
Canada	193,826	46,110	23.8	12,577	3,568	28.4	24.1
N.L.	12,685	2,515	19.8	432	111	25.7	20.0
P.E.I.	5,041	1,088	21.6	249	65	26.1	21.8
N.S.	10,256	2,966	28.9	439	136	31.0	29.0
N.B.	10,722	2,894	27.0	438	111	25.3	26.9
Que.	24,509	6,556	26.7	2,114	792	37.5	27.6
Ont.	51,758	12,632	24.4	4,067	1,004	24.7	24.4
Man.	16,001	4,096	25.6	619	205	33.1	25.9
Sask.	17,327	3,541	20.4	579	164	28.3	20.7
Alta.	21,841	4,772	21.8	1,526	401	26.3	22.1
B.C.	22,125	4,726	21.4	1,877	517	27.5	21.8
Y.T.	685	178	26.0	100	29	29.0	26.4
N.W.T.	634	137	21.6	85	29	34.1	23.1
Nvt.	242	9	3.7	52	4	7.7	4.4